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**A WEB-BASED GOVERNMENT CONTRACTS INFORMATION SYSTEM
WITH AUTHENTICITY VERIFICATION USING
STEGANOGRAPHY-BASED ENCRYPTION TECHNIQUE**

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The capstone project attached here to, entitled “**A Web-Based Government Contracts Information System with Authenticity Verification Using Steganography-Based Encryption Technique**” prepared and submitted by **Griffin L. De Leon, Charles Kristoffer A. Go**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology is hereby accepted.

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Introduction

In the digital age, the protection and authenticity of government documents have become critical concerns, particularly in procurement transactions where public funds are involved. Government procurement processes generate legally binding documents that must remain accurate, untampered, and verifiable. While procurement documents may undergo authorized revisions during drafting or formal amendment procedures, any unauthorized modification or post-approval alteration of finalized contracts undermines transparency, legal validity, and public trust. For instance, recent research on LGUs in the Philippines shows that weak records management and data security practices expose documents to manipulation (Tabia, 2025). Philippine jurisprudence also emphasizes that falsification of public documents by officials is a serious criminal offense that undermines institutional trust and accountability (Respicio, 2024; Res Picio, 2025). Studies further highlight gaps in LGU internal controls and governance systems that create opportunities for such wrongdoing (Juco, 2024). These findings indicate that relying solely on encryption may be insufficient document integrity and authenticity must also be proactively safeguarded, particularly in LGU settings.

Public procurement in the Philippines is governed by Republic Act No. 9184, also known as the Government Procurement Reform Act of 2003. This law establishes a standardized system designed to promote transparency, competitiveness, efficiency, and accountability in the acquisition of goods, infrastructure, and consulting services. Under RA 9184, government entities are required to follow a structured procurement cycle that includes procurement planning, preparation of the Approved Budget for the Contract (ABC), public posting of invitations to bid, submission and evaluation of bid proposals, issuance of a Notice of Award, contract signing, implementation, inspection, and final payment. Each stage mandates the

creation and retention of official records to ensure auditability and legal compliance. Oversight and post-audit functions are performed by the Commission on Audit, which may impose administrative sanctions for violations, including suspension, disqualification from bidding, and forfeiture of securities.

Because RA 9184 requires documentation at every stage of the procurement process, even small-scale local government units generate substantial volumes of records. Typical procurement documentation includes the Purchase Request, Approved Budget for the Contract (ABC), Request for Quotation or Invitation to Bid, Bid Proposals, Abstract of Bids, Technical Specifications, Notice of Award, Contract Agreement, Performance Security, Inspection and Acceptance Report, Disbursement Voucher, and supporting financial and delivery records. These documents collectively form the official procurement file and serve as the legal basis for disbursement and audit review. Public procurement is essential to government operations because it enables the acquisition of goods, infrastructure, and services necessary for delivering public programs and community development initiatives. At the barangay level, procurement facilitates the purchase of office supplies, equipment, repair works, and community project implementation. Without a functioning procurement system, basic administrative and development activities cannot proceed. Therefore, ensuring the integrity and authenticity of procurement documents is not merely a legal requirement but a fundamental component of effective local governance.

In the case of Barangay Sto. Niño, interviews with barangay officials reveal that procurement transactions occur approximately one to two times per month, totaling an estimated ten to thirteen procurement contracts annually. Each procurement contract averages sixty to seventy pages in length and is reproduced in multiple physical copies for distribution among the barangay, suppliers, contractors, and other participating parties. All procurement documents are

stored in physical filing cabinets, and retrieval occurs whenever prospective bidders, suppliers, contractors, or auditors request access for verification purposes. While this manual system complies with documentation requirements under RA 9184, it relies heavily on physical storage and visual inspection for authenticity verification.

The reliance on manual verification presents a potential vulnerability. Given the length of procurement contracts and the existence of multiple distributed copies, subtle alterations such as modification of financial figures, insertion of additional terms, or unauthorized edits to contractual clauses, may be difficult to detect without systematic integrity verification. Because of this effective records management is critical in procurement processes because each stage of the procurement cycle generates legally binding documentation that serves as the basis for financial disbursement and audit review. Proper organization, secure storage, version control, and controlled access to these records are essential to prevent discrepancies and ensure accountability. Weak document management systems increase the risk of lost records, undetected alterations, and audit findings, thereby undermining transparency and public trust. Although legal sanctions exist for falsification of public documents, detection mechanisms at the barangay level remain largely reactive rather than preventive. While digital editing tools make modification of PDF-based contracts increasingly accessible, current verification practices in the barangay setting do not incorporate tamper detection mechanisms.

To address this gap, this study proposes a web-based contract authentication system that integrates encryption and Least Significant Bit (LSB) steganography for PDF-based government procurement documents. The system embeds encrypted verification data within a digital seal image attached to procurement contracts, enabling detection of unauthorized modifications without altering the document's visual appearance. By combining cryptographic hashing, encryption, and steganographic embedding, the proposed framework enhances document

integrity verification while aligning with the transparency and accountability principles mandated under RA 9184. The system is designed to support secure digital transformation in local government procurement processes while preserving compliance with existing legal and regulatory requirements.

Research studies demonstrate that Least Significant Bit (LSB) steganography functions as an efficient method to embed data within digital files without modifying their visual appearance. The combination of LSB with encryption technology protects hidden data by both concealing it and providing encryption protection (Al-Shammari & Jassim, 2024). The availability of editing tools has made digital document forgery more accessible to users. The editing tools available today enable users to make subtle yet significant changes to PDF contracts and signatures and metadata which become difficult to detect through basic visual checks (Sharma & Singh, 2020). The current verification methods fail to detect these modifications so additional protection measures become essential.

Least Significant Bit (LSB) is categorized as a steganographic technique rather than a cryptographic algorithm. While cryptography (e.g., AES) transforms readable data into ciphertext to ensure confidentiality, steganography conceals the very existence of hidden data within a carrier file. In this study, AES is used to encrypt verification data, while LSB steganography is used to embed the encrypted data within the PDF's digital seal image. This layered approach ensures both confidentiality and concealment.

The research fulfills Sustainable Development Goal (SDG) 16 by supporting transparent governance systems with accountable institutions that serve all citizens. The research supports

SDG 9 through its development of technological solutions which enhance digital

infrastructure and improve document management systems in local government organizations. The research supports SDG 11 through its work on sustainable cities and communities because proper documentation of public procurement activities enables local projects to deliver their intended community benefits.

The research provides essential value to local governance digital transformation through its development of an advanced secure document management system with verification capabilities. The project implements steganography with encryption and web accessibility to defend government documents from tampering and unauthorized changes while supporting the development of trustworthy public institutions.

In Barangay Sto. Niño, procurement contracts are prepared manually by designated barangay personnel following the issuance of a Notice of Award. The contract document is typically drafted using standard word processing software, incorporating the terms specified in the bidding documents, approved budget allocation, scope of work, payment schedule, and compliance clauses mandated under RA 9184.

Once finalized, the contract undergoes review and approval by the Barangay Captain and authorized signatories. After signing, multiple physical copies are printed and distributed to the barangay office and the winning contractor. In some instances, a digital copy in Portable Document Format (PDF) is generated for archival or reproduction purposes. However, no cryptographic signature, hash verification, or embedded authentication mechanism is applied during document generation.

The contracts subject to this study are the finalized PDF versions of procurement agreements that contain complete contractual terms, financial allocations, and official signatures.

These documents serve as the primary legal basis for disbursement and audit review. Under the current workflow, authenticity verification relies solely on visual inspection and manual comparison of printed copies.

The proposed system intervenes after contract finalization and PDF generation. Prior to storage or distribution, the system embeds encrypted verification data within the contract's digital seal image using steganographic techniques. This ensures that any subsequent modification to the PDF file can be detected through integrity verification procedures.

While digital storage improves document preservation, verification of document authenticity remains dependent on access to the office computer where files are stored. During audits, contractor requests, or external verification, barangay officials must manually retrieve files to confirm legitimacy. This process becomes inefficient when verification is needed outside the barangay office.

With the increasing availability of smartphones and basic mobile literacy among barangay personnel, a mobile-based authentication tool can provide on-site verification of procurement contracts. Integrating a mobile scanning application with steganographic extraction and cryptographic validation mechanisms enables real-time contract authentication without requiring direct access to the office computer.

Background of the Study

Government agencies at the national and local levels maintain official records such as

procurement documents, contracts, financial reports, resolutions, and administrative certifications. These documents serve as the legal and operational basis for public spending and project implementation, ensuring transparency, accountability, and proper utilization of public funds. However, challenges in document management and verification continue to affect public institutions. According to Romero (2018), citizens often encounter delays, incomplete disclosures, and limited access when attempting to obtain procurement-related records, weakening public oversight and institutional trust.

Public procurement in the Philippines is governed by Republic Act No. 9184, also known as the Government Procurement Reform Act of 2003. RA 9184 establishes a structured procurement framework based on transparency, competitiveness, efficiency, and accountability. It prescribes a formal process that includes procurement planning, preparation of the Approved Budget for the Contract (ABC), public posting of invitations to bid, submission and evaluation of bid proposals, contract awarding, implementation, and post-procurement monitoring. Each stage generates official documents that must be preserved for audit and compliance purposes. The Commission on Audit (COA) conducts audits to ensure adherence to procurement regulations and financial accountability.

Despite this legal framework, implementation challenges persist across Local Government Units (LGUs). Querijero Roque and Santos (2024) identify unclear procurement policies, limited institutional capacity, and continued reliance on manual systems as contributing factors to delays and documentation irregularities. Santos and Gatchalian (2023) further emphasize that many barangays experience weak digital archiving systems, unsecured storage practices, and inconsistent document handling protocols. Similarly, Dela Cruz (2024) notes that barangays, as the smallest administrative units, often depend heavily on printed and scanned

documents for procurement and financial transactions due to limited digital infrastructure. The World Bank (2023) also observes that LGUs in developing regions frequently operate with fragmented or outdated information systems, increasing the vulnerability of official records.

The growing availability of digital editing tools has intensified risks related to document manipulation. Sharma and Singh (2020) highlight that modern PDF-editing software allows subtle yet high-impact modifications to financial figures, signatures, timestamps, and metadata without obvious visual traces. Tabia (2025) stresses that weak digital record-keeping practices in LGUs expose financial documents to a high risk of undetected manipulation, leading to discrepancies between declared and actual expenditures. Juco (2024) similarly points out that unreliable documentation undermines transparency in local project monitoring when alterations go unverified due to manual comparison techniques. These findings suggest that traditional visual inspection and physical signature validation may be insufficient to ensure document integrity in increasingly digital environments.

Various mechanisms exist to secure digital documents, including digital signatures, watermarking, Public Key Infrastructure (PKI), and forensic authentication techniques. While effective, these approaches may require specialized infrastructure, technical expertise, or formal digital certificate systems that are not fully implemented at the barangay level. Kozachok et al. (2022) and the EncCoD Project (2020) further explain that encryption alone, although essential for confidentiality, does not conceal the presence of sensitive data and may still attract targeted interception or cryptanalysis. These limitations highlight the need for additional layers of protection that enhance integrity verification without imposing complex infrastructure requirements.

In the case of Barangay Sto. Niño, procurement and contract documents are prepared and finalized following RA 9184 procedures. Records are stored physically in filing cabinets and verified through manual comparison of printed copies and signatures. While this system satisfies procedural requirements, it lacks tamper-proof digital authentication mechanisms. When contracts are reproduced or archived in PDF format, no cryptographic signature, hashing function, or automated integrity verification tool is applied. This creates potential exposure to unauthorized alterations, version inconsistencies, or falsified reproductions, particularly when documents circulate beyond the barangay office.

An interview in Barangay Sto. Niño also revealed that the barangay produces approximately 10–13 procurement-related contracts annually. These contracts are drafted in Microsoft Word using standardized templates and are finalized upon signature of the Barangay Captain and supplier. Digital copies are stored in a single office computer, while physical copies are archived separately. The barangay currently has no version control system or digital verification mechanism to detect post-approval alterations. Confusion over multiple revisions and difficulty detecting alterations in multi-page contracts were identified as recurring operational challenges.

Based on interviews conducted with the Barangay Captain and Secretary, digital copies of contracts are stored solely on the office computer, with no remote verification mechanism available. When contractors or auditors request validation of a contract's authenticity, verification requires manual retrieval and cross-checking of stored files. This limitation highlights the need for a portable verification system that can authenticate contracts outside the primary storage location.

Given these operational realities, integrating encryption with Least Significant Bit (LSB) steganography presents a practical enhancement to document security at the barangay level. Steganography provides a concealment-based layer of protection by embedding encrypted verification data within ordinary digital files (Patil & Jain, 2020). Least Significant Bit (LSB) steganography remains one of the most lightweight and practical techniques for this purpose, as it modifies the least significant bits of digital data without visibly affecting the document's form or usability (Sahu & Swain, 2021; Kaur et al., 2022). By embedding encrypted authentication information within finalized PDF contracts, the proposed approach strengthens document integrity verification while remaining aligned with existing procurement procedures under RA 9184.

Statement of the Problem

The Government Procurement Reform Act of 2002 through Republic Act No. 9184 requires that public procurement must be transparent, competitive, and accountable. Section 8 of RA 9184 explicitly mandates that “information and communications technology shall be utilized in the conduct of procurement procedures” to promote transparency and efficiency. Despite this mandate, many barangays continue to rely on manual and semi-digital systems in handling procurement documents.

Based on the interview conducted with the Barangay Captain of Barangay Sto. Niño, procurement documents such as Purchase Requests, Requests for Quotation, Abstract of Canvass, Purchase Orders, and Contract Agreements are primarily prepared using printed templates, signed physically, and stored in filing cabinets. Digital copies, when available, are saved in standalone computers or shared folders without structured verification mechanisms. Authenticity checking is performed manually through physical signature comparison, which is

time-consuming and highly dependent on human judgment. This approach increases the risk of undetected alterations and loss of records, especially in cases of file duplication or physical damage.

Supporting this concern, the Commission on Audit (COA) Annual Audit Reports (2023) documented several barangays where procurement documents lacked proper authorization signatures, contained inconsistencies between approved and actual purchased items, and in some cases showed signs of altered entries submitted to justify expenses. These findings demonstrate how the absence of secure verification mechanisms creates opportunities for document tampering and falsification.

Furthermore, a study published in the International Journal of Research and Creative Media Studies (IJRCMS, 2024) emphasized that although the legal framework encourages ICT adoption in procurement, “significant gaps remain in actual implementation,” particularly in cybersecurity integration and data-protection protocols. These gaps leave digital documents as vulnerable as paper-based records.

1. Lack of Secure and Centralized Digital Verification System in detecting unauthorized modification or verifying document authenticity.

Despite the ICT mandate under Section 8 of RA 9184, many barangays including Barangay Sto. Niño do not operate a unified digital verification platform capable of detecting unauthorized modifications to procurement contracts. Verification is conducted manually, relying on visual inspection and signature comparison, which does not guarantee document integrity once copies are reproduced or distributed. The COA (2023) reports of altered procurement records further illustrate the consequences of lacking a

centralized authentication mechanism. Without a secure and structured verification system, document authenticity cannot be reliably validated.

2. Limited Access and Transparency for Public Oversight due to Inefficient Utilization of ICT in Procurement Processes.

RA 9184 promotes transparency and public participation through electronic posting and monitoring of procurement activities. However, inefficient utilization of ICT tools limits public accessibility. As observed in Barangay Sto. Niño, procurement contracts are not available through a structured online platform that allows residents to independently verify authenticity. This situation aligns with findings from IJRCMS (2024), which noted that weak implementation of digital procurement systems reduces transparency and weakens independent monitoring. When citizens cannot easily access or verify official contracts, oversight mechanisms remain dependent on internal administrative processes.

3. Absence of Advanced Security Mechanisms for Digital Documents

Even when procurement documents are digitized, advanced security protections such as encryption, integrity fingerprinting, and embedded authentication markers are not commonly implemented. RA 9184 mandates ICT utilization but does not specify the technological safeguards required to prevent document falsification. IJRCMS (2024) highlighted that weak cybersecurity integration and insufficient data-protection protocols make electronic procurement documents susceptible to unauthorized alteration. Without

encryption and tamper-detection mechanisms, digital contracts may appear visually legitimate despite having been modified, undermining public confidence and institutional accountability.

Objective of the Study

The objective of this study is to design, develop, and implement a web-based application that enhances the security of government PDF contracts through the integration of encryption and Least Significant Bit (LSB), providing multi-layer protection for sensitive information. Specifically, the system aims to address key security concerns and achieve the following Objectives:

Specific Objectives

1. To design and implement a web-based application/system that can extract and embed encrypted data within PDF documents.

This objective addresses the lack of a secure and centralized digital verification system by developing a platform capable of generating document-specific integrity data and embedding encrypted authentication markers within procurement contracts. The system utilizes the PyPDF library for extracting document metadata and content structure, and the Django framework for managing the application's core processes and interface.

2. To design and implement a structured web-based public access and verification portal that enables transparent viewing and independent authenticity checking of barangay procurement contracts.

By establishing a centralized online platform where authenticated procurement contracts can be viewed by authorized users and verified independently by residents. The system provides a public-facing interface that supports document search, structured presentation, and authenticity validation without allowing modification, thereby strengthening transparency, accountability, and ICT utilization in compliance with RA 9184.

3. To apply Least Significant Bit (LSB) steganography for embedding encrypted data within PDF files while integrating an encryption technique to further enhance data security.

The study implements a LSB technique to embed encrypted data within PDF files without affecting the document's structure or readability. This approach enhanced the overall security of the file by concealing sensitive information within its binary data, providing an additional layer of protection against unauthorized access and tampering. Furthermore, a multi-layer security mechanism is introduced through the integration of the AES encryption algorithm, which strengthens data confidentiality by encrypting the information prior to the embedding process. While integrating a unique integrity fingerprint on each document based on the formula given. This will also be treated as a secret key that will be stored through our database.

Rationale of the Study

In Barangay Sto. Niño, procurement contracts are prepared following the procedures mandated under Republic Act No. 9184. The barangay processes approximately one to two procurement transactions per month, totaling around ten to thirteen contracts annually. Each

contract averages sixty to seventy pages and is reproduced in multiple physical copies for distribution to the barangay office and the contractor. These documents are stored in filing cabinets and retrieved manually whenever verification is required by suppliers, bidders, or auditors.

Accountability in the current system is maintained through signature authentication, review by authorized officials, and controlled physical storage. However, integrity verification relies primarily on visual inspection and manual comparison of distributed copies. When contracts are converted to PDF format for reproduction or digital archiving, no cryptographic signature, hashing mechanism, or tamper-detection system is applied. Given the length and complexity of procurement contracts, subtle alterations may be difficult to detect without systematic digital verification.

The proposed system enhances the existing accountability framework by embedding encrypted authentication data within finalized PDF contracts using steganographic techniques. Rather than replacing current administrative procedures, the system strengthens them by introducing automated integrity validation that complements manual controls. For Barangay Sto. Niño, where procurement activities occur regularly throughout the year, the implementation of this mechanism provides scalable protection for all procurement-related documents generated annually.

Although digital signatures and Public Key Infrastructure (PKI) provide strong authentication mechanisms, their implementation requires certificate authorities, structured key management, and dedicated infrastructure. Based on the operational conditions observed in Barangay Sto. Niño, where procurement documents are drafted using standard office software

and stored in a single office computer without centralized digital certificate management, implementing a full PKI-based system may not be practical at this level. Therefore, this study explores a lightweight steganographic approach combined with encryption that aligns with the barangay's technical capacity.

Least Significant Bit (LSB) steganography was selected because it enables authentication data to be embedded within an existing PDF seal image without requiring external certificate systems or specialized hardware. Compared to visible watermarking, LSB preserves the document's visual appearance, which is important for maintaining the official format of procurement contracts. This makes LSB a practical and scalable solution aligned with the technical capacity and operational workflow of the barangay.

The localized application of this solution directly contributes to the United Nations Sustainable Development Goals. Under SDG 16 (Peace, Justice and Strong Institutions), the system strengthens institutional integrity by protecting procurement documents from unauthorized modification and supporting transparent governance. Under SDG 9 (Industry, Innovation and Infrastructure), the implementation of encryption and steganography in a web-based authentication system demonstrates the use of innovative digital infrastructure to enhance public sector operations. Under SDG 11 (Sustainable Cities and Communities), improved document authenticity supports responsible monitoring of local projects and public resource utilization.

For local government officials, the system provides a practical safeguard against unauthorized document changes while maintaining compliance with RA 9184 transparency

requirements. For citizens, it enables access to procurement documents with greater confidence in their authenticity, thereby promoting participatory governance. For the Commission on Audit or COA, the system offers an additional verification reference that may support document validation during audit reviews.

Given that contracts are stored digitally without version tracking and are frequently requested for audit and contractor verification purposes, ensuring document integrity becomes operationally critical. The absence of automated authenticity verification mechanisms increases the risk of unnoticed alterations, especially in multi-page contracts that undergo several revisions prior to finalization.

This study strengthens document integrity at the barangay level by integrating cryptographic and steganographic verification into existing procurement workflows. By enhancing authenticity assurance while preserving accessibility and transparency, the system contributes to improved governance, accountability, and public trust.

The integration of a mobile-based verification tool strengthens the system's practicality by enabling contract authentication beyond the confines of the barangay office. This feature supports audit readiness, contractor transparency, and remote verification while maintaining data integrity through embedded cryptographic mechanisms.

Scope and Delimitations of the Study

The study focuses on the design and implementation of a Web-Based Government Contract Information System with Authenticity Verification that enables the public, particularly barangay residents, to view verified copies of government contracts, while allowing authorized barangay personnel to upload, manage, and authenticate procurement-related documents such as

Purchase Requests (PR), Requests for Quotation (RFQ), Abstract of Canvass, Notice of Award, Purchase Orders (PO), and Contract Agreements for barangay-funded projects. The primary goal is to ensure that public records remain accessible, genuine, and protected against tampering or unauthorized alteration. The system was developed specifically for Barangay Sto. Niño, Biñan City, based on concerns identified during the interview with its Barangay Captain, including the heavy reliance on physical documentation, lack of digital verification mechanisms, and limited familiarity with government procurement standards under Republic Act No. 9184 (Government Procurement Reform Act). By digitizing contract handling, the system provides a modernized and secure workflow for storing, managing, and verifying specific procurement and contract documents commonly produced by the barangay, including PRs, RFQs, Abstract of Canva ss, Notice of Award, Purchase Orders, and Contract Agreements.

The system incorporates a multi-layer security framework, which includes the document's integrity established through the generation of a Canonical Fingerprint (CF), which maps its internal structure and components to create a unique digital identity. This fingerprint, along with supplementary verification data, is then hardened using AES Encryption to prevent unauthorized access or tampering. To maintain the document's visual fidelity, these encrypted markers are covertly embedded into the PDF via Least Significant Bit (LSB) Steganography. This technique integrates the security data at the binary level, ensuring the authenticity markers are effectively invisible while leaving the original content entirely undisturbed.

These layered security methods allow the platform to detect unauthorized modifications in the document's text, images, and structural elements, enabling it to distinguish genuine contracts from altered or falsified versions.

The contracts covered in this study include annual procurement and project-based

agreements produced by Barangay Sto. Niño, such as supplier contracts, infrastructure-related agreements, and service contracts. These contracts are typically prepared in Microsoft Word using official templates, finalized upon signature of the Barangay Captain and supplier, and stored in both physical and digital formats. Given that these documents often consist of multiple pages and undergo several revisions prior to finalization, the absence of version control and digital verification mechanisms increases the risk of undetected alterations.

System Features Included in the Scope

The system includes role-based user access control that differentiates permissions between the Barangay Administrator and Clerk to ensure proper account-based authorization. It provides a secure contract upload mechanism and protected document storage environment for procurement-related records. A public contract viewing interface is made available for barangay residents to promote transparency while maintaining document integrity. The platform incorporates automated authenticity verification and tampering detection mechanisms, supported by an activity log or audit trail that records contract access and verification attempts for accountability purposes. The system also includes a web-based administrative platform responsible for contract upload, encryption, steganographic embedding, and secure storage. In addition, a mobile-based authentication application is developed to scan digital or printed contract copies and extract embedded steganographic verification data. The mobile application is strictly limited to authenticity verification and does not permit contract modification, editing, or uploading functions.

Limitations of the Study

The scope of this study is limited to the barangay level, particularly Barangay Sto. Niño, Biñan City. While the Commission on Audit (COA) is acknowledged as an important oversight institution in validating government contracts, the study does not include COA as a direct participant or research locale. Instead:

This study recognizes COA's oversight role but focuses solely on the barangay's internal processes and implementation.

The system does not support contract workflows at the municipal, provincial, or national government levels, as its implementation is confined to Barangay Sto. Niño. It is limited to PDF-based barangay procurement documents and may not fully accommodate complex or non-standard document formats. Public users are restricted to viewing authenticated contracts and are not granted privileges to upload, edit, or modify records. The system does not prevent document misuse once a copy has been downloaded outside the platform environment. Furthermore, authentication capabilities apply only to contracts processed through the proposed system and do not extend to externally generated documents. The platform does not replace existing procurement procedures or bidding processes mandated under Republic Act No. 9184. The mobile application functions solely as a verification tool and does not operate as a comprehensive document management system.

Conceptual Framework

This is the conceptual framework of the proposed Web-Based Government Contracts Information System, which addresses the current system's restrictions in terms of data authenticity, confidentiality, and integrity through steganography-based encryption techniques as

shown in Figure 1. A robust digital solution would boost efficiency, transparency, and fraud prevention in public procurement. These gaps indicate basic information needs for government oversight bodies, contractors, and the public. A Secure and Verifiable Document Repository is needed to ensure that government contracts are stored in a tamper-proof, secure, and integrity-preserving environment. The extremely limited public access to information also requires efficient searching and retrieval of procurement records. Most importantly, there is a need to protect documents from unauthorized alterations more effectively than traditional methods.

The needs of the three participants must be addressed in the project design and implementation for the system to succeed. The existing solutions employed in the management of government contracts exhibit several limitations. Access to procurement documents is restricted, and few mechanisms exist to independently verify their authenticity. Additionally, basic digital signatures do not adequately protect documents from forgery or unauthorized replication. Furthermore, contract verification is manual, relying on human validation, which is error-prone, time-intensive, and lacks an immutable audit trail. Such conditions are contrary to the objectives of Republic Act No. 9184 (RA 9184), which aims to ensure better quality and safer procurement processes through transparency and accountability.

To address these limitations, a complete encryption and decryption mechanism is integrated into the document verification process. When a contract document is uploaded into the system, the encryption procedure begins by extracting all textual, image, and vector components from the PDF file. These extracted elements are normalized to ensure consistent formatting and encoding. After normalization, each component undergoes hashing using the SHA-256 cryptographic hash function. The resulting hashes are combined together with relevant document

metadata such as file name, timestamp, author, and version number. This combined data forms a Canonical Fingerprint (CF), which serves as a unique and irreversible representation of the document's exact content at the time of upload.

Once the Canonical Fingerprint is generated, it is secured using symmetric encryption. An HMAC is first applied to provide integrity authentication, ensuring that any modification to the fingerprint can be detected. The fingerprint is then encrypted using the AES-256-CBC algorithm to guarantee confidentiality. For secure key management, the AES encryption key may be wrapped using RSA public-key encryption, thereby implementing a hybrid cryptographic approach that combines symmetric efficiency with asymmetric security.

After encryption, the protected Canonical Fingerprint is embedded into a high-resolution 300 DPI PNG or BMP representation of the document using Least Significant Bit (LSB) steganography. The encrypted payload is inserted into the least significant bits of selected pixels, ensuring that the visual appearance of the document remains unchanged and imperceptible to the human eye. This steganographic embedding binds the encrypted fingerprint directly to the document itself, creating a tamper-evident mechanism that travels with the file.

During verification, the decryption process is initiated. The system extracts the hidden LSB payload from the uploaded document image and decrypts it using the corresponding RSA private key to unwrap the AES key. The AES-256-CBC algorithm is then used to decrypt the Canonical Fingerprint, and the HMAC is validated to confirm integrity. Simultaneously, the system re-extracts and normalizes the current document contents and generates a new SHA-256-based fingerprint following the same canonicalization procedure. The newly computed fingerprint is then compared with the decrypted original Canonical Fingerprint. If the two

fingerprints match exactly, the document is verified as authentic and unaltered. If any discrepancy is detected, the system flags the document as tampered or modified.

Through this structured encryption and decryption process, the system ensures confidentiality by protecting sensitive procurement data, integrity by detecting any unauthorized alterations, and authenticity by verifying that the document originates from a legitimate and unchanged source. This mechanism strengthens compliance with Security Theory principles and supports the transparency and accountability goals mandated by RA 9184.

The implementation of this system is expected to create significant opportunities for positive transformation in government procurement. Public trust and transparency will be strengthened as citizens gain access to verifiable contract documents, reducing suspicions of corruption, misinformation, or manipulation. The digitalization and automation of contract storage, retrieval, and verification will streamline government workflows, improving accessibility and reducing administrative burdens. Finally, fraud, tampering, and unauthorized alterations are expected to be significantly mitigated through the advanced steganography-based encryption mechanism, reinforcing document integrity and providing a reliable defense against malicious attacks.

Diagram:

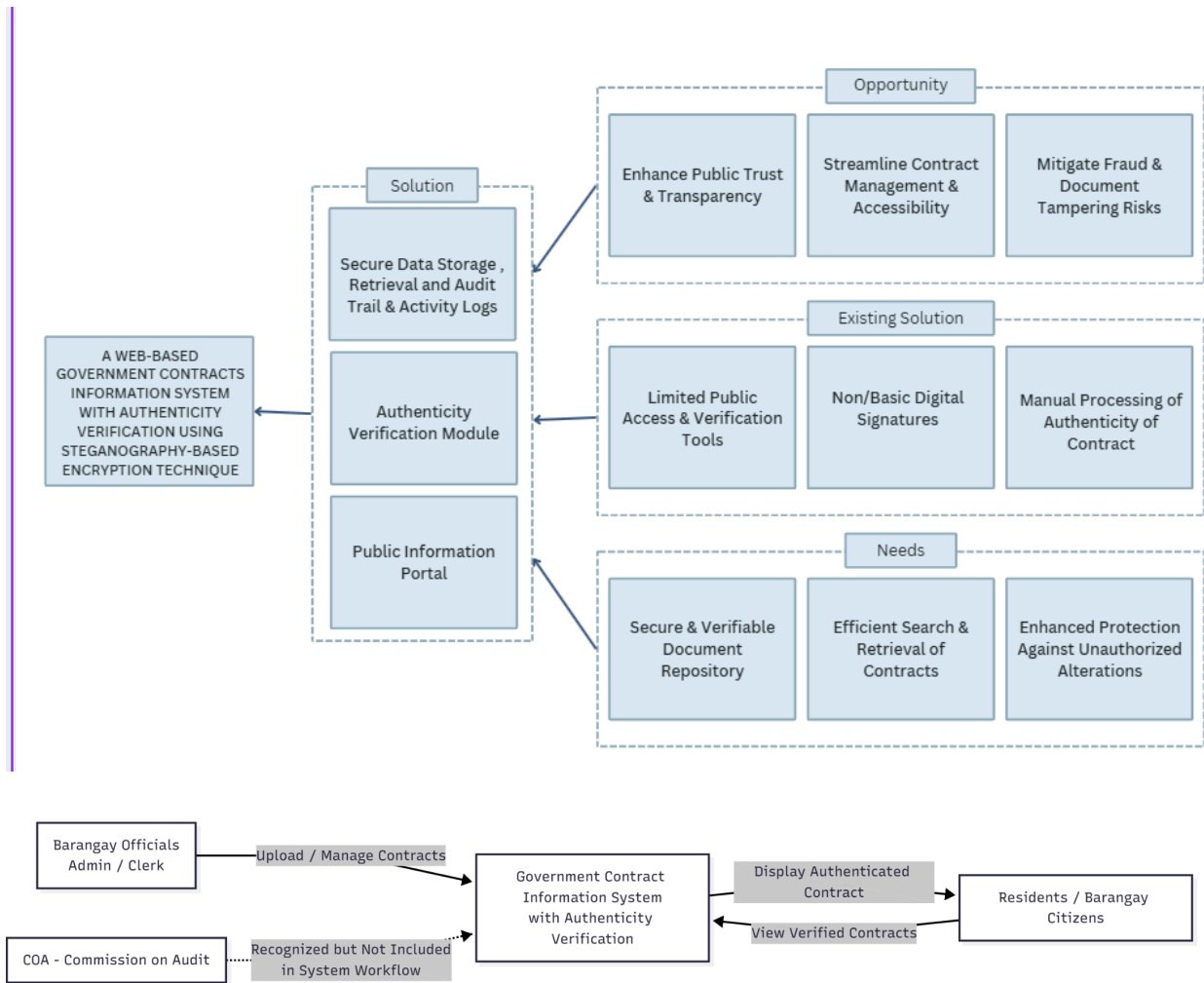
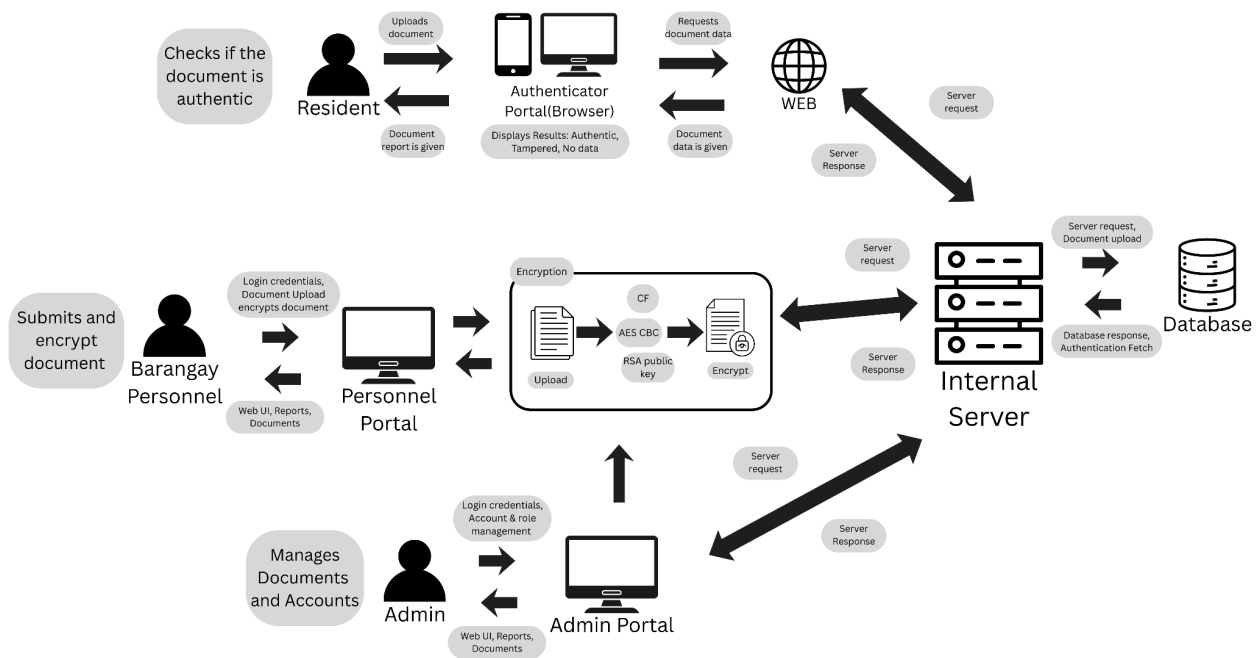


Figure 1.0 Conceptual Framework of the Study

System Architecture



The image shows a secure document authentication and verification system architecture illustrating how residents, barangay personnel, administrators, servers, and the database interact to validate document authenticity.

On the left side, a Resident uploads a document through an Authenticator Portal (browser-based) to check whether the document is authentic. The portal sends a request to the web server, which forwards it to the internal server. After processing, the system returns verification results such as Authentic, Tampered, or No Data, which are then displayed to the resident.

Barangay personnel use a Personnel Portal to submit and encrypt official documents before storing them in the system. During upload, the document undergoes a hybrid encryption process, combining AES (CBC mode) for document encryption and RSA public key encryption for secure key handling. This ensures the integrity and confidentiality of stored records.

An Admin Portal is shown at the bottom, where administrators manage accounts, roles, and documents. Admins oversee system operations and ensure that only authorized users can access or process sensitive information.

At the center-right is the Internal Server, which handles server requests and responses. It processes authentication checks, verifies encrypted document data, and communicates with the Database. The database stores encrypted document records and authentication data. When a verification request is made, the internal server retrieves the stored encrypted data, performs validation checks, and sends the result back through the web layer to the user.

Arrows in the diagram represent bidirectional communication between:

- User portals and the web layer
- Web layer and internal server
- Internal server and database

Overall, the image presents a secure document authentication and verification framework where uploaded documents are encrypted, stored securely, and later verified through controlled server-side validation to ensure authenticity, integrity, and protection against tampering.

Definition of Terms

I. Cryptography and Security Concepts

AES (Advanced Encryption Standard) – A symmetric block cipher used to encrypt and decrypt data using a shared secret key. This study utilizes AES-256 in Cipher Block Chaining (CBC) mode.

Canonical Fingerprint (CF) – A unique hash-based digital identifier that represents the

full content of a document, combining textual, visual, and metadata elements. In this study, the CF is used for authenticity verification.

CBC (Cipher Block Chaining) – A mode of operation for AES encryption in which each plaintext block is XORed with the previous ciphertext block before encryption.

Data Encryption Key (DEK) – A symmetric key used to encrypt and decrypt data within AES encryption.

Decryption – Process of restoring encrypted data to its original readable form with a key.

Digital Certificate – An electronic credential issued to confirm the identity of an entity or system involved in secure communications.

Digital Signature – A cryptographic technique that verifies the sender's identity and ensures that a document has not been altered, providing integrity and non-repudiation.

Encryption – The process of converting readable data into an unreadable format to protect it from unauthorized access.

File Hashing – The process of generating a unique hash value from a file to verify its integrity during upload or verification.

Hashing Algorithm – A mathematical function that transforms input data into a fixed-length hash value, ensuring that even minor changes in the data result in a completely different output.

Least Significant Bit (LSB) – A steganographic method that hides data in the least significant bits of an image or file without visibly altering it.

Public Key – A cryptographic key used to encrypt data in asymmetric encryption systems, accessible to anyone.

SHA-256 (Secure Hash Algorithm 256-bit) – A cryptographic hash function that produces a 256-bit unique digital fingerprint of data, used in this study to verify

document integrity and detect tampering.

Steganography – The practice of concealing secret data within another digital medium, such as an image or document. In this study, it is applied through the LSB technique to embed encrypted document hashes within a digital seal.

II. System and Information Management Concepts

Access Control — A security mechanism that restricts user privileges based on their assigned roles or permissions.

Algorithm — A defined sequence of computational steps used to perform a specific function such as encryption, decryption, or verification.

Audit Trail — A chronological record of user actions and system events that ensures accountability and supports transparency in document transactions.

Authorization — The process of determining which actions or data a verified user is permitted to access.

Information System — A coordinated structure of people, software, and hardware designed to collect, process, store, and distribute information.

Metadata — Descriptive data about a file, such as author, date of creation, modification timestamp, file size, and embedded identifiers. In this study, metadata is included in the Canonical Fingerprint for verification purposes.

User Authentication — The process of verifying a user's identity before granting access to system resources.

Web-Based System — An application that operates on a web server and is accessed by users through a web browser.

III. Governance and Legal Framework Concepts

Authenticity Verification — The process of confirming that a document originates from a

legitimate source and has not been altered since its issuance.

Document Integrity — The assurance that a document's content remains accurate, complete, and unaltered throughout its lifecycle.

E-Governance — The application of information and communication technologies to improve the delivery of public services, transparency, and efficiency in government operations.

Local Government Unit (LGU) — A political subdivision responsible for administering a specific area, such as a barangay, municipality, or city.

RA 9184 (Government Procurement Reform Act) — The Philippine law that establishes guidelines for public procurement, promoting transparency, accountability, and fair competition in government transactions.

Review of Literature

Topic Section

Government Procurement and Document Management

Government Procurement Contracts

Studies of e-procurement systems demonstrate clear operational benefits for public procurement. An example is the article by Yumang and Mantillas (2025) on the procurement processes in barangays of the City of Manila, which found that while compliance with Republic Act 9184 (RA 9184) was moderate, significant challenges remained such as lack of training, low budgets, and lengthy processes which hindered efficient procurement operations.

Republic Act No. 9841

Republic Act No 9184, also known as the Government Procurement Reform Act was enacted in 2003 to standardize and modernize procurement processes in the Philippines. The law seeks to promote transparency, accountability and efficiency in government transactions by institutionalizing the use of information and communications technology in all stages of procurement. According to De Guzman (2022) the implementation of this law significantly improved the procurement landscape by introducing the Philippine government electronic procurement system or PHILGEPS for short, which gave increased public access to procurement information, allowing the public to monitor how public funds are spent, De Guzman also noted that while the law established a strong legal foundation for transparency, its success is dependent on the capability of local government units (LGU) and as such Yumang and Mantillas (2025) observed that while barangay level procurement units demonstrate moderate compliance

to this law, they still rely on manual or semi digital processes, due to their limited training, lack of digital infrastructure and weak document security hinder their full compliance to the Republic Act of 9184. The proposed system aims to address one of the Act's implicit challenges, verifying the authenticity of electronic procurement documents, through this the study aims to contribute to continuing modernization efforts envisioned by the Republic Act 9184.

Government Procurement Systems

The use of digital tools in buying stuff for government work made things more clear, cutting down chances for shady deals. Back in 2005, Siar found out that when purchasing is done by machines, every step can be tracked so people involved see how contracts move along right away. On top of that, the COA said in 2022 that online-based buying slashed repeated paperwork, sped up checking records, while also pushing local areas to report in a uniform way.

Online Contract Approval and Monitoring Systems:

According to the research, the introduction of PHILGEPS reduced manual processing and paperwork, centralized data management, and simplified contract approvals by providing an online bidding and monitoring environment. The study found that this digital platform made procurement information more accessible to the public, promoting accountability and minimizing corruption risks. However, it also revealed several limitations, including inconsistent internet access, lack of technical training among government employees, and the need for better verification mechanisms to authenticate electronic documents (UST–Legazpi, 2015).

Document Management Systems for Government Use:

Modern document management systems (DMS) in public offices aim to reduce reliance on paper-based filing and improve record retrieval efficiency. Dela Cruz (2024) found that web-based DMS solutions used by selected municipalities in Laguna enhanced collaboration between departments and reduced administrative delays. Despite these improvements, the study warned that weak encryption and lack of authenticity verification mechanisms made sensitive documents vulnerable to unauthorized alterations. This finding supports the integration of cryptographic and steganographic methods for secure document handling in local government systems.

Document Management Systems (DMS) at the LGU level

Research on web-based DMS deployments in Philippine local government units shows benefits in retrieval and collaboration but flags weak encryption, incomplete integrity checks, and low ICT skills among staff as recurring problems. These weaknesses point to the potential value of adding document concealment/authentication layers to existing DMS implementations. World Bank. (2023)

Information System

The field of Information Systems (IS) addresses how organizations capture, process, store, and distribute information using technology. A recent scoping literature review titled Information systems function research: a scoping literature review and research agenda examines the role of the IS function (ISF) in organizations, highlighting emerging opportunities, challenges and the evolving importance of digital transformation Veloso, M., & Varajão, J. (2025).

Information Systems in Public Service

Tahir along with Lamakaraka (2023) take a close look at past studies on Management Information Systems within public administration, showing that when MIS works well, people tend to get better government services. Even though these systems can seriously boost how smoothly agencies run, actual rollout hits snags - like lack of funds, skill gaps, or resistance inside organisations, which keeps benefits out of reach. The research brings solid evidence into conversations about tech use in public work, pointing out how far behind reality lags from expectations when it comes to using MIS in state-run operations.

Government Procedures on Public Rights of Information

Government procedures for public rights of information outline how citizens request, access, and appeal for documents and data held by the state. Literature reviews highlight procedural inefficiencies (delays, high fees, vague exceptions), the role of digital portals for open data, and the importance of clear guidelines and enforcement to make rights of information meaningful in practice (KPMG Philippines, 2021).

Philippines Rights of Information

The right of access to information in the Philippines is framed by public records laws and freedom-of-information initiatives, which require government agencies to provide citizen access to documents, subject to exceptions. Literature explores challenges in implementation: red-tape, inconsistent agency responses, lack of digital systems, and low public awareness. Improving transparency relies on technology, policy and institutional culture. Senate of the Philippines (n.d)

Authenticity Verification and Security in Government Documents

Authenticity Verification

Checking if something's real means making sure it hasn't been changed, comes from a proper place, and matches what it claims to be - whether it's info, ID, or files. Elsbach and Bhattacharya (2020) say being genuine happens when inner traits line up with outward behavior, showing honesty and reliability. Lehman and team (2022) broke this down more, pointing out four aspects - how steady it is, how well it fits norms, personal links, and ongoing presence that shape whether things feel truly legit in different situations. When dealing with tech or data-heavy environments, proving something's valid keeps shared or sent details trustworthy; Chen, Chen, and Xie (2021) worked on ways to test just how truthful social media content really is. On another front, Nikulchev and others (2021) used these checks on web-based questionnaires, figuring out if answers came from actual people instead of bots. Beyond just numbers, O'Connor's team in 2022 showed how it helps tell real online feedback apart from made-up comments; meanwhile, media researchers at Fiveable point out that checking facts is key to staying trusted when sharing news. In total, confirming what's authentic works across fields - linking honest people, secure digital spaces, and dependable tech - to block lies and keep info networks reliable.

Authenticity Verification for Philippine Government

Beyond checking who someone really is in person, the Philippines set up ways to confirm if digital papers are real to make online services smoother while cutting down fake docs. The PSA rolled out electronic versions of records like births, weddings, and deaths; each comes with a special code users can check live on the web (PSA, 2023). At

the same time, the DFA started using e-Apostilles slapping verified state paperwork with encrypted tags plus scannable squares before sending them overseas (DFA, 2024). People abroad can prove validity simply by snapping the QR symbol or typing the ID into [Apostille.gov.ph](https://apostille.gov.ph) - a quick way to know it's legit and from an approved source. Still, right now, this first round only includes personal status forms handled by the PSA, so broader use will have to wait. These efforts show how the Philippines is slowly taking stronger steps toward trustworthy online governance. Though issues still exist like helping people everywhere learn digital skills and improving internet access building better tools to check documents on the web moves things forward without relying on paper, while also reducing scams in today's tech-driven world.

Online Document Verification Frameworks

Recent frameworks for online document verification (including approaches based on self-sovereign identity and cryptographically verifiable credentials) demonstrate practical architectures for web-based verification systems that do not leak private data while enabling public checks of authenticity. Such frameworks inform design decisions for a public authenticator that verifies whether a PDF was legitimately issued by a barangay without exposing hidden content. (Satybaldy, Subedi, & Nowostawski, 2022).

Hybrid cryptographic methods for authenticity:

Hybrid approaches (encryption combined with other verification techniques) have been shown to strengthen document origin and integrity guarantees. For government documents, combining cryptographic signing/verification with additional concealment

mechanisms can reduce both tampering and unauthorized visibility (Alanzy, Alomrani, Alqarni, & Almutairi, 2023).

Techniques used to Prevent Forgery and Tampering Government Documents

Techniques used to prevent forgery and tampering include embedding digital watermarks or steganography, using tamper-evident formats, applying digital signatures and timestamping, implementing document version control and audit trails, and establishing verification APIs for public and internal checking. Studies show that integrating cryptographic hashing with logging and public-key infrastructure strengthens document integrity and authenticity (Prashant, Haque, Kaur, & Yadav, 2024).

PDF fake news/forgery

In 2018, the office of the ombudsman reported that a number of barangay officials were ordered jailed for forging barangay resolutions, falsifying documents and holding sham meetings to illegally appoint officials. According to the ombudsman's decision, the officials fabricated resolutions by falsely recording that a council meeting took place, listing members as "present" even though no such meeting was held. The fabricated resolution included manually altered signatures.

The investigation revealed that the barangay officials had deliberately altered the resolution's contents to make it appear legitimate, taking advantage of the fact that barangay documents are often kept only in printed or scanned form without verification features. Once the falsified resolution was digitized and circulated as a PDF, it became nearly indistinguishable from authentic barangay records. This enabled the fraudulent

document to pass through several layers of local administration before the inconsistencies were discovered. This shows that even the smallest unit of local governance, document forgery can occur through simple alterations to signatures, timestamps and narrative content. Such vulnerabilities underscore the necessity of implementing secure verification procedures, to ensure integrity and prevent unauthorized alterations.

Security Protocols for Government Documents

The Philippines keeps data safe using clear rules set by law and daily operations, making sure info stays private, accurate, and reachable when needed. Office of the President Memo Circular 78 from '64 lays out how papers should be labeled split into Top Secret, Secret, Confidential, plus Restricted with each level showing how much damage leaked details might cause to national safety. This order spells out steps like tagging files right, sending them in two envelopes, while keeping them locked up tight or inside areas off-limits to most people. Getting near these documents depends on needing that specific knowledge at that time - not just anyone can look - even if you have clearance fit for certain levels. These steps make up the heart of how officials handle sensitive papers, keeping openness while protecting country secrets. On top of that, there's a system for paper handling tracking who grabs files, plus ways to shred secret documents so no one digs them back up.

To keep digital info safer, the 2012 Data Privacy Act (RA 10173) requires every govt agency acting as a PIC to use management practices, real-world protections, besides tech tools when handling private or delicate data. Instead of weak options, the National Privacy Commission pushes strong encoding like AES-256 for saving and sending files, according to their memo NPC Circular No. 16-01. Because of these steps, personal

records stay shielded against leaks, hacking attempts, or changes done without approval. On top of that, DICT's GIS Plan covering 2022 through 2028 sets out rules on who can view certain materials, demands locked-down transfer channels, along with reliable login setups. When combined, all these actions show how the Philippines handles docs using old-school secrecy levels together with up-to-date privacy tactics so both citizen and state data remain protected.

Technical Foundations of Secure Digital Systems

Encryption Algorithms for Securing Files

Recent work in file encryption demonstrates that securing documents—for example contract-PDFs or official records—requires more than simply applying standard encryption. Banga et al. (2023) developed a file encryption/decryption framework using AES for both text and image files, emphasising performance and integrity of large-file sets. Ryu (2023) proposed a hybrid approach combining legacy ciphers with AES-256 to improve resistance to sophisticated attacks on files. Additional studies, such as Musa (2023), evaluate the real-world suitability of different algorithms for securing files, including considerations of algorithm complexity and resource constraints. Taken together, these works show that encryption algorithm choice, key management, and implementation context (such as file size, storage medium, transmission channel) are critical when securing file-based documents.

Encryption and Cryptography(Overview & Encryption Process):

Crypto relies on math along with clear rules to protect data, ensure correctness, make signatures solid. Encoding information typically involves selecting an approach,

reusing one key for large pieces, switching to separate keys if sending hidden messages or applying signatures, while also generating and safeguarding these keys carefully, leaning on trusted methods like AES-GCM, handling their life cycle via changes, deletions, swaps. Recent checks measure pace, difficulty in key control, strengths across environments - AES vs RSA vs ECC - with safe coding habits helping builders match tools to website file security demands. Kshetri, Rahman, Rana, Osama, & Hutson (2024)

Comparison on Digital Document Security Encryptions Techniques

Comparative literature on digital document security and encryption techniques examines symmetric versus asymmetric cryptography (e.g., AES vs RSA/ECC), authenticated encryption modes (GCM, CCM), key distribution and management, performance trade-offs, and implementation contexts (cloud, mobile, archives). Those reviews help researchers decide which cryptographic methods best suit particular document security use-cases (Yin et al., 2022)

Steganography and Data Hiding

Artz (2001) provides a foundational overview of digital steganography, describing it as the art of inconspicuously embedding information within ordinary data such as images, audio files or text, without altering unintended recipients to its existence, it has also noted the difference between encryption and steganography, encryption makes data unreadable while steganography simply hides the data

Steganography on PDF

Tyagi, Dwivedi, and Saxena (2020) discussed that PDF documents, due to their platform independence and wide use, are ideal carriers for information hiding. Their method focused on embedding secret messages within text-based PDFs using imperceptible coding transformations that preserve document readability. The study

concluded that PDF text steganography enhances secure document transmission by achieving both high hiding capacity and strong imperceptibility. Koptyra and Ogiela (2020) observed that steganographic techniques can be effectively applied to PDF files by modifying internal document structures such as dereferenced objects and page trees. Their study demonstrated that hidden data can remain completely invisible in standard PDF readers while maintaining the document's normal appearance. The research concluded that structural steganography in PDFs provides a secure and high-capacity medium for covert communication, especially when combined with content-based hiding methods.

Steganography-Based Encryption Techniques

LSB (Least Significant Bit) Steganography

Patil, P., & Jain, R. (2020) A review on image steganography techniques concluded that the Least Significant Bit (LSB) approach remains one of the most efficient and lightweight methods for concealing data. Their findings emphasize that LSB manipulation can embed information within digital files without noticeably altering their appearance or quality

Reinforcing LSB, Singh, S., & Attri, V. K. (2015) proposed a dual layer mechanism that integrates AES encryption with LSB steganography. Ensuring the even if encrypted data is intercepted, it remains concealed within a cover medium, this research demonstrated combining these two techniques improves data protection and resistance to tampering,

Hidden Data Techniques for Document Authenticity

Hussain and Muhammad (2023) introduced a digital text-watermarking

approach that embeds invisible identification codes within PDF documents to verify originality and detect post-creation alterations. Their findings revealed that hidden watermark markers can remain intact even after common document modifications, enabling organizations to validate the legitimacy of official records and prevent undetected tampering. This study demonstrated that embedding concealed watermark data within documents provides a reliable layer of integrity verification alongside traditional security controls.

Blue, Condell, and Lunney (2024) applied steganographic data-embedding techniques to identity and legal documents to ensure non-repudiation and forgery resistance. Their work highlighted that embedding hidden verification codes within the document structure allows authenticity checks without altering readability or format. By concealing validation keys within the document itself, this method enhances tamper-resilience and reduces the risk of fraudulent duplication or the circulation of falsified official records.

Digital Signature and Verification

Sharma and Bhattacharya (2022) reviewed various digital signature mechanisms used to verify the authenticity of electronic documents, including RSA-based and hash based schemes. Their analysis revealed that signatures not only confirm document ownership but also ensure data integrity by detecting unauthorized modifications.

Rashid and Khan (2025) conducted a comparative study on the deployment of digital signatures across individual and organizational contexts, in which they

emphasized in the importance of public key infrastructure in establishing trust and compliance with legal frameworks

Titov et al, (2022) further explored this within contractual and legal processes, suggesting a functional symmetry approach that ensures signatures, verification and document lifecycle management remain consistent across systems.

Commission on Audit (COA, 2021) also released formal guidelines on the use of electronic and digital signatures in philippine government transactions.

Hybrid Cryptographic Approaches

Hybrid cryptographic methods for authenticity:

Hybrid approaches (encryption combined with other verification techniques) have been shown to strengthen document origin and integrity guarantees. For government documents, combining cryptographic signing/verification with additional concealment mechanisms can reduce both tampering and unauthorized visibility (Alanzy, Alomrani, Alqarni, & Almutairi, 2023).

Balhaf et al, (2025) designed a hybrid cryptographic system that merges RSA encryption with Least Significant Bit (LSB) steganography to embed ciphertext within image files, This integration made extraction by unauthorized parties significantly more difficult Eswaran et al. (2025) expanded upon this concept by combining public key cryptography with steganographic embedding in multimedia files.

A similar model by Satybaldy, Subedi, & Nowostawski, (2022) introduced an

online document verification framework using self-sovereign identity and verifiable credentials.

A 2024 MDPI study on cloud data security presented a four step hybrid encryption process that integrates the use of AES-256, RSA and Steganography to ensure the integrity and security of sensitive files.

Integration of Steganography and Cryptography

Almuhammadi and Al-Shaab (2013) observed that using cryptography or steganography alone will be insufficient for security, however combining these two can create a more reliable and stronger system. The study concluded that Digital Signature and Verification Methods Furthermore, Subsequent research has further validated this hybrid method. For instance, Singh and Attri (2015) introduced a dual layer mechanism that integrated AES encryption with LSB steganography, showing that even if encrypted data is intercepted, it remains concealed within a cover medium.

Similarly, Balhaf et al. (2025) designed a hybrid cryptographic system combining RSA encryption and steganographic embedding, enhancing both confidentiality and authentication during transmission.

The encryption ensures that only authorized users can decrypt the document while steganography enables authenticity verification by embedding hidden identifiers or watermarks directly within the document file, mitigating risks and unauthorized modification of official records

Web-Based System Architecture and Methodologies

Web Application Security Best Practices

Web applications that expose document upload/verification APIs must follow secure coding and API security guidelines (OWASP Top 10 / API Top 10) especially strong authentication, proper access control, input validation, and secure transport to prevent tampering, replay, and unauthorized access to verification services. These recommendations are essential when building a public authenticator and a private encryption module.

Information System Security in E-Governance

E-governance systems face unique security challenges because they collect, store, and process large volumes of sensitive citizen and institutional data while providing wide public access. Recent literature emphasizes a layered approach (policy, technical controls, and continual monitoring) and aligning controls to standards (e.g., NIST) to maintain availability, confidentiality, and integrity while managing regulatory and privacy requirements. Studies also highlight the need for resilience against increasingly sophisticated attacks and the importance of governance, risk management, and compliance (GRC) frameworks in public-sector deployments (Singh & Karaulia, 2011).

Implementation of Secure File Transmission in Web System:

Sending files safely online usually relies on methods that scramble commands and information together like SFTp, FTPS or API setups built on HTTPS. Experts suggest going with login keys instead of passwords, managing servers and crypto-keys carefully, checking data hasn't been altered with things like HMACs or secure encryption and setting tight usage limits, keeping close logs, plus scanning devices for malicious software, Most split setups mix encrypted communication layers, say SFTP over TLS, together with app-level safeguards such as one-time upload tokens, automatic virus

checks and locking down each file individually according to Bob Ertl(2025)

Data Integrity and Confidentiality in Public Systems:

In public setups, making sure data stays correct and secure usually depends on several layers - encryption hides details, whereas tools like digital signatures or MACs confirm nothing's been altered. Entry gets closely restricted, yet records show every action taken by users. Rules such as those in NIST SP 800-53 outline different steps aimed at state groups and large orgs; following these builds trustworthy protections that are both reviewable and law-compliant. JTF (2020)

Database Management Systems

A database manage system serves as the backbone of numerous information systems as they provide structured storage, retrieval and manipulation of data while also enabling data organization and security even with multiple users at the same time. (Elmasri & Navathe, 2017), Database Management Systems can reduce redundancy and improve consistency by enforcing data relationships through schemas and constraints, so that stored records can remain accuracy to their source and synchronized across modules.

Firebase Realtime Database, the database that will be used in this study's proposed system, operates as a cloud-hosted NoSQL database that supports real-time data synchronization and secure authentication features. According to Google Cloud (2023) firebase ensures security through server-enforced rules and integration with firebase authentication, which regulates user permissions and protects stored data from unauthorized access, in terms of our proposed system this is a good fit, firebase manages encrypted PDF contracts and their metadata, while role-based access control ensures that only verified barangay staff can upload, modify and verify the records.

Role-Based Access Control

Role-Based Access Control (RBAC) is a security mechanism that restrict system access to authorized users based on their assigned roles within an organization, instead of assigning permissions per user, RBAC associates privileges with roles that reflect their job responsibilities RBAC simplifies permission management while maintaining compliance with organizational policies (Ferraiolo & Kuhn, 1992).

Sandhu, Coyne, Feinstein, and Youman (1996) introduced one of the most influential models of RBAC, which outlined how role hierarchies and constraints can efficiently manage permissions across large scale systems. RBAC is applicable because administrative roles inherently map to permissions in system workflows. For example, each official has distinct duties: approvals vs. document uploads vs. record-access. Aligning system permissions the system can then simplify access management and secure sensitive contract documents.

Agile Methodology

Agile methodology emphasises iterative development, frequent delivery of functional increments, close stakeholder collaboration, and adaptive planning. In the context of software and systems development, Agile has been credited with improving responsiveness to change, reducing time-to-market, and enhancing user satisfaction. However, literature also points out challenges: scaling Agile in large organisations, maintaining documentation and governance, and balancing flexibility with control (Daraojimba, Nwasike, Adegbite, Ezeigweneme, & Gidiagba, 2024)

Web Design and UI/UX Principles

Web Design plays a vital role in the usability, accessibility and overall success of web based systems, According to Krug (2014) a well designed website should only prioritize its simplicity and clarity, so users can navigate and perform their designated tasks without being confused by its layout, you can achieve this through consistent layout, clear visual hierarchy and intuitive interface elements, Garrett (2011) also mentioned that it's not merely about its design but also about creating a meaningful interaction between the user and the system, ensuring that design decisions align with users' needs and goals.

Web systems that adhere to government or institutional functions must also adhere to these design principles. As Marcotte (2011) introduced, responsive web design enables websites to automatically adjust layouts and elements across various devices and screen sizes, so that the user experience is consistent across numerous devices in the context of our system, these web designs ensure that barangay staff and the public can easily access and verify the documents without any confusion or hassle.

Synthesis

The reviewed literature collectively highlights the continuing challenges and opportunities in modernizing document management and procurement processes within the Philippine local government context, studies on government procurement and RA 9184 consistently emphasize the importance of transparency, accountability, and efficiency in public transactions, yet reveal persistent barriers like lack of technicality, manual procedures and the lack of security mechanisms (Uumang & Mantillas, 2025; De Guzman, 2022) which help support the development of our proposed system.

Literature on Document Management Systems and Information systems in public administrations

demonstrate that web based solution can help streamline operations (Dela Cruz, 2024; World Bank, 2023), however, the recurring weakness in their encryption, verification and user training highlights the need for security focused enhancements.

The studies on authenticity verification and hybrid cryptographic frameworks further strengthen this direction by proving that the combination of encryption, hashing and concealment methods can significantly enhance document integrity and security (Alanzy et al., 2023; Satybaldy et al., 2022), The philippine government's recent moves toward digital authentication such as PSA's e-certificates and DFA's e Apostilles mean it aligns with secure and verifiable e governance practices to our proposed system (PSA, 2023; DFA, 2024).

Research on web application security and secure file transmission underscores the need for layered security measures based on standards like OWASP and NIST to mitigate threats from unauthorized access and tampering (Singh & Karaulia, 2011; OWASP, 2023), multiple of these studies collectively validate the adoption of AES-256, SHA-256 and the LSB steganography that will be used for the core mechanisms of the proposed system.

As well as insights from the UI/UX design, Agile Methodology, and database management system emphasize the usability, adaptability as crucial elements in a system's success (Krug, 2014; Garrett, 2011; Google Cloud, 2023) together these works can give us a foundation for the proposed system's study giving a technical perspective for development.

CHAPTER 3

Methodology

This presents the methodological function for the development, evaluation and validation of the proposed system, the methods were structured to ensure the system addresses the core problems identified in the barangay's current workflow such as lack of secure verification, manual workflows, vulnerability to tampering and limited accessibility.

The methodology begins with the research design being quantitative, which provides objective measurements for the system's accuracy, security and user's feedback, This is reinforced via through major evaluative components, System Accuracy Testing which measures the reliability of encryption, canonical fingerprint, authentication and verification, and the User Acceptance Testing (UAT) which assesses the usability, functionality and overall readiness of the system.

The research locale, Barangay Sto. Niño, Biñan City, was selected due to its reliance on manual documentation processes and evident challenges in verifying government records. By focusing on an LGU with real operational constraints it helps ensure our study is technically feasible and relevant.

Central to the methodology would be the Agile Development Methodology, which supports iterative development, from the designing, planning and developing modules including authentication, encryption and decryption, LSB-based steganography, document storage and the tools used to verify documents. It also details the technical framework, incorporating AES-256 CBC, encryption SHA-256 hashing for canonical fingerprint generation and least significant bit (LSB) steganography for concealed authenticity embedding within PDF documents. Which also help identify the gaps identified under RA 9184. Finally, an outline of the data gathering, including document review and Testing sessions, The data analysis plan establishes how

quantitative results from Accuracy tests, UAT scores and verification outputs will be evaluated to determine the system performance and acceptance.

3.1.0 Research Design

The study will utilize a **Quantitative Research Design** to evaluate the performance and acceptance of the barangay management system, The quantitative approach is appropriate as Creswell (2018) emphasized, quantitative research enables the collection of measurable data that can be analyzed statistically to draw objective conclusions for the system accuracy and usability which allows the measure of performance, like how accurate the authentication is or how secure the encryption is or the overall experience with the system. Such metrics include the accuracy of generated data and the User Acceptance Testing (UAT) ratings of system respondents.

System Accuracy Testing will focus on evaluating the reliability of the system, in the processing, the encrypting and the authentication of the data, it will measure how accurately the system performs the intended functions compared to the expected results.

User Acceptance Testing (UAT): It will involve the intended end-users, executing a predefined set of critical business tasks. Which will determine if the system meets the functional requirements and is suitable for the live environment.

Combined findings of both the user acceptance testing and system accuracy testing can offer a comprehensive view of the system's reliability, which can serve as a foundation for interpreting results in the data analysis plan. The proposed system also incorporates a multi layer encryption and verification framework composed of the following key algorithms and cryptographic processes of AES-256 Encryption Key along with Cipher Block Chaining (CBC) mode, SHA-256 Hashing Algorithm and LSB Steganographic Embedding key.

3.2.0 Research Locale/Respondents

The study will be conducted in **Biñan City, Laguna**, specifically in **Barangay Sto. Niño** where the barangay office will serve as the testing and information-gathering site. The locale was selected because it represents a highly urbanized and rapidly developing local government unit (LGU) facing contemporary challenges in records management and public service delivery

Barangay Sto. Niño is situated in the landlocked component city of Biñan, Laguna, located in the CALABARZON region (Region IV-A), approximately 35 kilometers south of Metro Manila. Biñan City itself is renowned as a major commercial and trading hub south of the capital, which means a high level of economic activity. According to Philatlas (2020) Sto. Niño is highly accessible, being near major transport routes like the National Highway and the South Luzon Expressway (SLEX). Its proximity to commercial centers like Pavilion Mall further underscores its role as a busy, accessible administrative center within the city. its environment also provides a crucial testbed to ensure that the proposed system is not only technically sound but also implementable. According to data obtained from their Barangay Secretary as of 2025, Barangay Sto. Niño has an estimated population of 12,303 residents, meaning they experience administrative processes daily that's related to barangay operations, making it an appropriate test environment for the system's performance

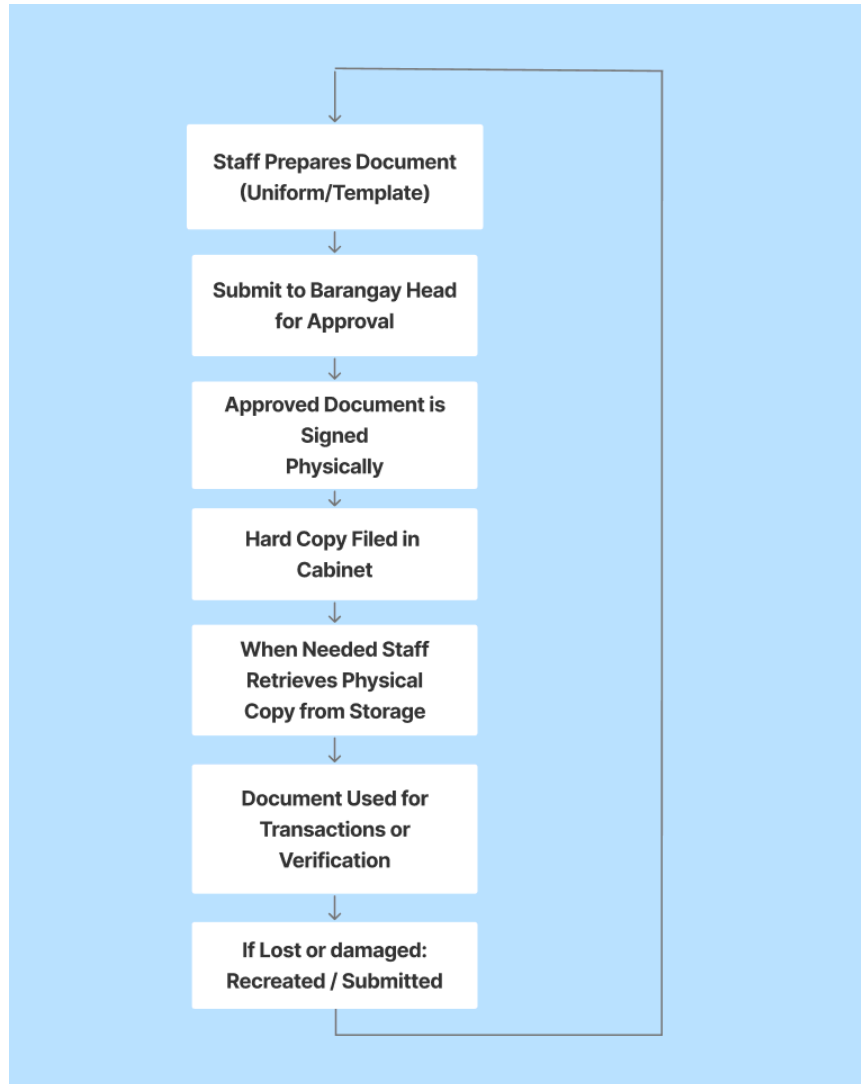


Figure 3.0 Barangay's Current Workflow

At present, Barangay Sto. Niño utilizes a manual document management process. Procurement records are prepared using standardized printed templates, physically signed by authorized officials, and stored in filing cabinets within the barangay office. Document verification is conducted manually through visual comparison of physical copies and handwritten signatures.

While this workflow satisfies procedural compliance under RA 9184, it presents several security

and authenticity vulnerabilities. Because documents are validated solely through physical inspection, subtle alterations—such as modification of financial figures, replacement of pages, or insertion of additional clauses—may go undetected. Physical signatures confirm approval but do not guarantee that the document content has remained unchanged after signing.

For users and requesters such as auditors, contractors, and citizens seeking verification, authenticity is dependent on trust rather than technical validation. There is no cryptographic mechanism to confirm that the retrieved copy is identical to the originally approved contract. In cases where documents are lost, damaged, or recreated, there is no mathematical baseline or integrity reference to determine whether the reproduced version matches the original.

The reliance on physical storage also exposes records to risks such as misplacement, unauthorized access, environmental damage, and delayed retrieval. Additionally, the absence of digital backups increases vulnerability in the event of disasters. These limitations result in reduced traceability, susceptibility to unauthorized alterations, and reliance on manual verification processes that are prone to human error.

The primary target users of the proposed system include barangay officials—such as the Barangay Captain, secretary, treasurer, and administrative staff—who are directly involved in document preparation, approval, and safekeeping. Citizens and external requesters who require document verification will also benefit from improved authenticity assurance mechanisms. Approximately thirty (30) citizens will serve as test users during User Acceptance Testing to evaluate whether procurement documents can be authenticated efficiently and whether tampering can be reliably detected.

Formal permission to conduct data gathering was obtained through a signed letter of request

approved by the Barangay Captain. Respondents for system evaluation will be selected using stratified sampling to ensure representation across organizational roles during User Acceptance Testing and cybersecurity evaluation.

3.3.0 Software Methodology

The study will utilize the Agile Development Methodology as the software development framework, Agile was chosen as it supports, iterative design, continuous user feedback, and adaptiveness, which are essential for developing systems intended for real world use in barangay operations, especially since barangay operations often involve evolving workflows and policy adjustments.

The development process will be divided into several sprints, each representing a short and focused development cycle that delivers a working component of the system as based on our sprint schedule.

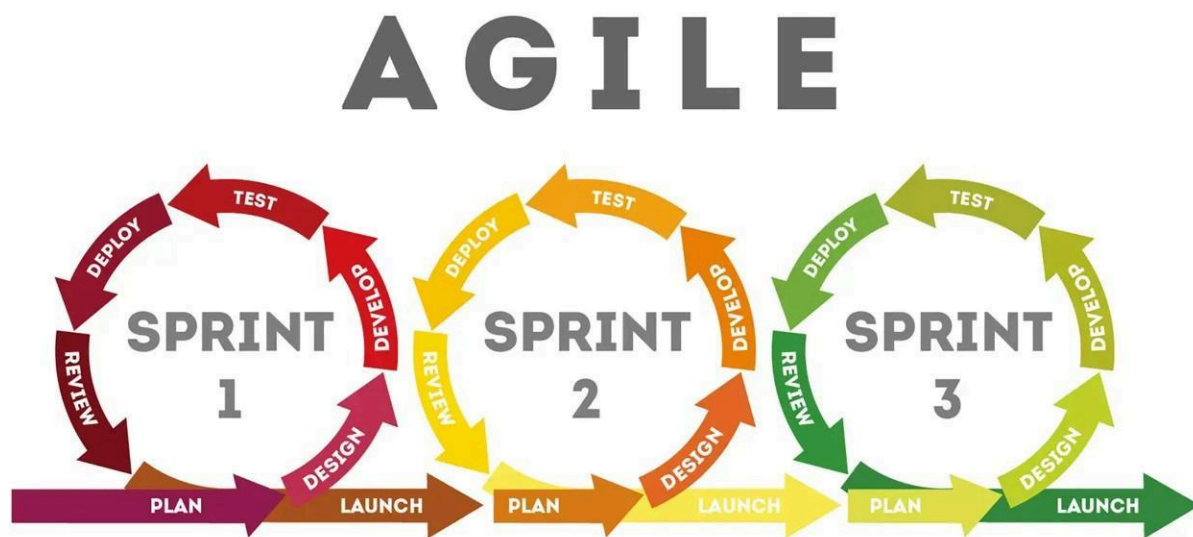


Figure 3.1 Agile Methodology Model Diagram

The process shown on Figure 3.1 provides quick responses to client feedback to create the project's definite direction during the software development cycle Arsenault (2017) has stated in contrast to traditional web development methodology, like the Waterfall model that uses traditional sequential approaches. Agile provides a non-sequential approach since there will always be technological bugs and new technical limitations to overcome, it also improves communication between the users and the development team utilizing the power of modern technology with a per-sprint-meeting feedback collection.

3.3.1 Seven Phases of the model:

- **3.3.1.1 Planning Phase**

The planning phase will serve as the foundation of the system's development, focusing on the project's scope, identifying requirements and aligning all activities with Republic Act 9184, ensuring that the researchers have a clear understanding of the barangay's workflow before introducing technological interventions.

The researchers conducted an assessment of the current document handling and procurement process within Barangay Sto. Niño, Biñan, Laguna, in which they rely heavily on manual filing and physical storage of contracts, resolutions and procurement related documents. This baseline analysis revealed issues such as delayed document retrieval and absence of a standardized system for verifying documents, These observations then guided the definition of requirements for the proposed system.

3.3.1.1.1 Identification and rationale alignment

Consultations were carried out with the barangay captain and staff to determine their needs and expected benefits from the system, the rationale of the project was to assist the barangay in complying with transparency and accountability standards that were mandated by RA 984 and Commision on Audit (COA), The proposed system would also benefit residents who wish to verify uploaded public documents such as barangay contracts and procurement records so they can ensure no unauthorized modifications occur.

3.3.1.1.2 Technical Requirements

During this phase the researchers defined the core technical components of the system. This included the Encryption and Decryption mechanisms which are the Advanced Encryption Standard (AES) with Cipher Block Chaining (CBC) mode and SHA-256 hashing to ensure the integrity of the document, each uploaded document will generate a canonical fingerprint, which serves as a unique validation key for authenticity verification. Authentication via Role Based Access Control (RBAC) ensuring that only authorized personnel can approve or verify documents as well as a data flow diagram to outline how documents will move through the system, from the uploading, encryption, draft.

In addition to the web-based system, the researchers defined requirements for a mobile-based authentication application. The mobile application will function as a portable verification tool designed for on-site or remote contract authentication. It will allow authorized users to scan digital or printed procurement contracts and extract embedded steganographic data for validation.

The mobile application must be capable of:

- Extracting LSB-embedded encrypted fingerprints from the contract's digital seal
- Decrypting AES-protected hash values
- Recomputing the document's SHA-256 hash
- Comparing extracted and recomputed values
- Displaying a clear authentication result (Authentic / Tampered)

This addition ensures that contract verification is no longer restricted to the barangay office computer.

3.3.1.1.3 Project Scheduling and Deliverables

An agile sprint schedule was developed to visualize the project timeline and duration of each phase under the Agile Development methodology. Each sprint focuses on a specific module such as user authentication, encryption process, file storage and verification.

3.3.1.2 Design Phase

The design phase will focus on translating the gathered requirements and insights from the planning phase into concrete system specifications. Researchers will begin conceptualizing the system architecture and creating mockups that represent both the administrative and public modules of the proposed system such as a database schema, data flow diagrams, and entity relationship diagrams as well as mockups to represent user interfaces. The design will also define the role-based access structures based on the organization hierarchy. Main objective being to ensure the system's design aligns with the data gathered through the planning phase.

During the design phase, a separate but interconnected mobile authentication module was conceptualized. The mobile application architecture was designed to interact with the same cryptographic framework used by the web system.

The mobile application consists of:

1. Scanning Interface Module
2. Steganographic Extraction
3. AES Decryption Module
4. Result Display Interface

The design ensures that the mobile application operates strictly as a verification tool and does not allow document modification, upload, or deletion.

To further reinforce this phase, records such as their procurement charts, organizational charts will be used to guide the creation of realistic data and role-based access structures within the system. Which can help determine which users will have privileges, who can upload and verify and how they will be categorized. Which will also give us insight regarding the design of our system architecture.

This phase will also involve the creation of the system's core algorithms which define how major functions operate, Including file encryption, user authentication and verification. The researchers will also develop the algorithm to handle the secure upload of document files as it is essential to ensuring that data are processed accurately, securely and in accordance with the system's intended workflows

3.3.1.2.1 Algorithm Design

To ensure the system's data integrity and security the researchers designed two core algorithms, one algorithm will handle the document encryption and uploading, while the second governs the verification process, Together it ensures that the files when retrieved or stored will remain authentic, confidential and free from tamper.

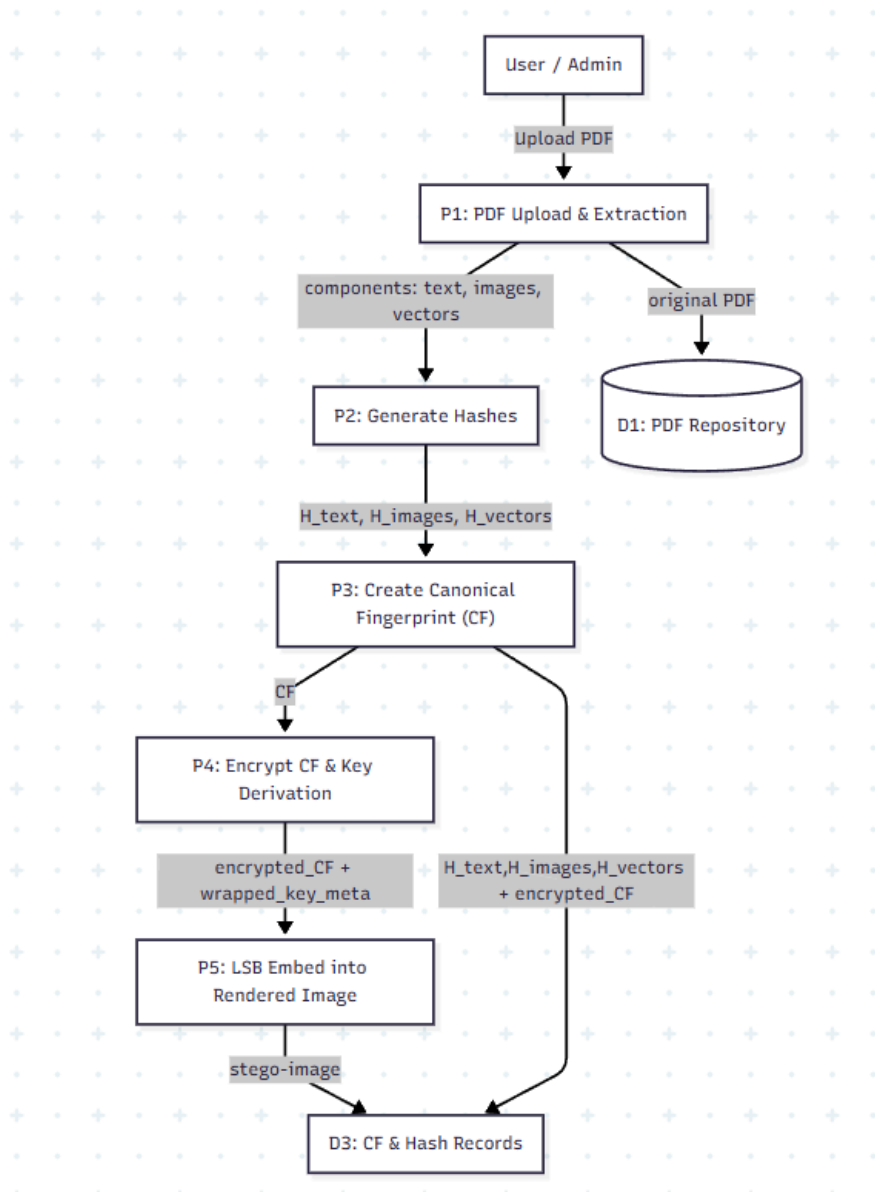


Figure 3.2 document upload and three layer encryption flow

```

BEGIN
    // P1: Upload and Extraction
    user ← Authenticate(User_Credentials)
    IF user IS NOT verified THEN
        DISPLAY "Unauthorized access"
        EXIT
    END IF

    extracted_components ← Extract(PDF_file, components=[text, images, vectors])
    Store(original PDF_file) INTO PDF_Repository

    // P2: Generate Hashes (Layer 1: Integrity)
    H_text ← SHA256(extracted_components.text)
    H_images ← SHA256(extracted_components.images)
    H_vectors ← SHA256(extracted_components.vectors)

    // P3: Create Canonical Fingerprint
    Canonical_Fingerprint ← Combine(H_text, H_images, H_vectors, Timestamp, User_ID)

    // P4: Encrypt CF and Key Derivation (Layer 2: Confidentiality)
    AES_Key ← Generate_Key()
    Encrypted_CF ← AES_Encrypt(Canonical_Fingerprint, AES_Key)
    Wrapped_Key ← Encrypt_Key(AES_Key, System_Public_Key)
    Metadata_Record ← {Wrapped_Key, File_ID, Owner_ID, Timestamp}

    // P5: LSB Embedding (Layer 3: Authenticity)
    Digital_Seal ← Generate_Seal_Template()
    Stego_Image ← LSB_Embed(Digital_Seal, Encrypted_CF)

    // Upload to Database
    Upload({
        PDF_file,
        Stego_Image,
        Metadata_Record
    }) TO CF_Hash_Records_Database

    DISPLAY "Upload Successful. Document Encrypted and Stored Securely."
END

```

Figure 3.3 document upload and encryption Pseudo Code

The following figure 3.1 as well as the figure 3.2 pseudo code illustrates the algorithm for document encryption and uploading, the process begins when the user selects a document for upload, the system then generates an encryption key using AES-256, and a hash value using SHA-256 of the encrypted file, and it is computed and embedded into a digital seal image using the least significant bit (LSB) steganographic technique, This seal, together with the encrypted document will be uploaded to the

database along with metadata such as file ID, owner and timestamp, the algorithm ensures that only verified users can perform uploads.

The encryption process implements a three layer encryption model, consisting of: (1) SHA-256 hashing for ensuring that data has not been altered, (2) a digital seal generated through LSB steganography for authenticity validation, and (3) AES-256 encryption for document confidentiality. A cryptographic key is also generated and wrapped as metadata to ensure only authorized users can decrypt the encrypted canonical fingerprint

To enable portable authentication while preserving structural integrity validation, the system introduces a Digital Seal mechanism. The Digital Seal functions as a visible authentication marker placed within the contract document while simultaneously serving as a carrier of hidden cryptographic data.

The Canonical Fingerprint (CF), generated from the combined hash values of textual content (h_{text}), images (h_{images}), and vector elements (h_{vectors}), is first encrypted using AES-256 encryption. The encrypted CF is then embedded into the Digital Seal image using the Least Significant Bit (LSB) steganographic technique.

Visually, the Digital Seal appears as a standard official marker (e.g., a barangay verification stamp). However, at the pixel level, selected least significant bits contain the encrypted structural identity of the document. This layered design ensures that:

1. The document maintains human-readable authenticity through the visible seal.
2. The embedded CF preserves cryptographic identity.
3. Any digital modification affecting the document structure will invalidate the

stored fingerprint.

This approach transforms the seal from a simple visual marker into a cryptographic authenticity carrier.

The implementation of LSB in this study is not a direct textbook embedding of arbitrary data. Instead, it forms part of a layered integrity verification framework. Prior to embedding, a Canonical Fingerprint (CF) is generated to represent the structural identity of the PDF contract. This fingerprint undergoes AES encryption before being embedded within a controlled image region of the document. The embedding process is restricted to non-structural components to prevent corruption of the PDF format. This approach demonstrates the application of cybersecurity principles such as confidentiality, integrity assurance, controlled attack surface, and layered defense.

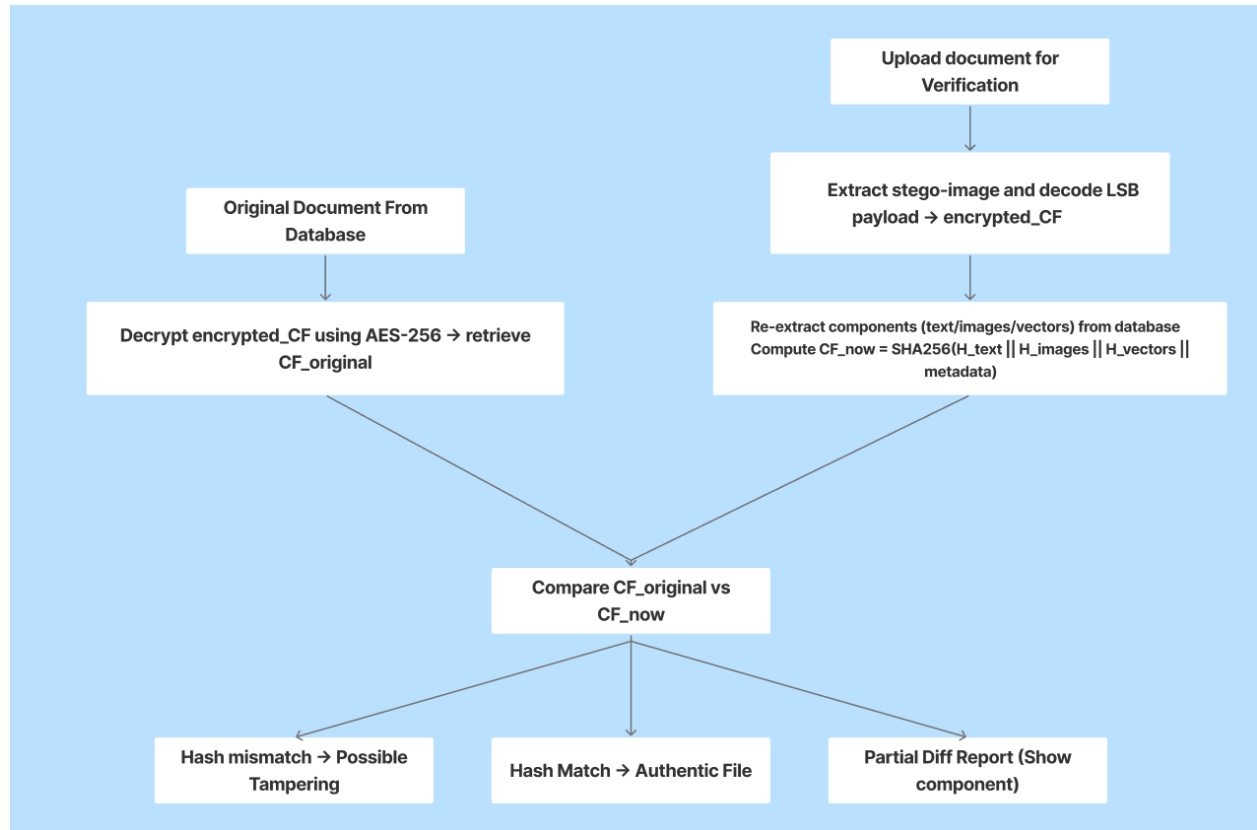


Figure 3.4 Document Verification flow

```

Begin
    // Step 1: Retrieve Original Encrypted Fingerprint and Metadata
    encrypted_CF ← database_record.CF_encrypted
    wrapped_AES_key ← database_record.metadata.wrapped_key
    stored_metadata ← database_record.metadata

    // Step 2: Unwrap AES Key using System Private Key
    AES_key ← Decrypt_Key(wrapped_AES_key, System_Private_Key)

    // Step 3: Decrypt the Canonical Fingerprint
    CF_original ← AES_Decrypt(encrypted_CF, AES_key)

    // Step 4: Extract Components from Uploaded Document
    extracted_text ← Extract_Text(uploaded_document)
    extracted_images ← Extract_Images(uploaded_document)
    extracted_vectors ← Extract_Vectors(uploaded_document)
    extracted_metadata ← Extract_Metadata(uploaded_document)

    // Step 5: Compute Canonical Fingerprint for Uploaded File
    H_text ← SHA256(extracted_text)
    H_images ← SHA256(extracted_images)
    H_vectors ← SHA256(extracted_vectors)
    CF_now ← Concatenate(H_text, H_images, H_vectors, extracted_metadata)

    // Step 6: Compare Canonical Fingerprints
    If CF_original == CF_now Then
        VerificationStatus ← "Authentic File"
    Else
        // Step 7: Generate Partial Difference Report
        diff_components ← CompareComponents(CF_original, CF_now)
        VerificationStatus ← "Possible Tampering Detected"
        OptionalDiffReport ← GenerateDiffReport(diff_components)
    EndIf

    // Step 8: Return Results
    Return (VerificationStatus, OptionalDiffReport)
End

```

Figure 3.5 Document Verification Pseudocode

The following figures ensures and compares document authenticity by comparing the hash value extracted from the embedded seal image with the hash stored in the database, if both hash values from the document in the

database and the document in the user's hands match, the document can be considered authentic, otherwise it is flagged with possible tampering.

For the key, the system retrieves the wrapped AES key stored in the document's metadata. This key will then be decrypted using the key's private data

These algorithms help the system's backend logic during the development phase, particularly in database integration and encryption modules.

For the key, whenever a verification request happens: The system fetches the stored wrapped_key from the database.

- **3.3.1.3 Development Phase**

Development phase will focus on translating the approved system design and specification into a functional prototype of the system, guided by the agile methodology, the phase will be divided into multiple sprints. Each lasting approximately one to two weeks.

The system will be developed using a dual-platform approach consisting of a web-based administrative system and a mobile-based authentication application.

The web-based system will be developed using HTML, CSS, JavaScript, Django (Python), and Firebase for backend and database services.

The mobile authentication application will be developed using (insert

your platform — Android Studio with Kotlin / Flutter / React Native). The mobile application will implement image processing and steganographic extraction mechanisms compatible with the encryption framework used by the web system.

During this stage the researchers will code the system in order for it to be compatible with the development environment established during the previous two phases. The system will be developed using a modular approach, which presents four core modules of the system: (1) Report Dashboard, (2) User Management and Authentication (3) Encryption and Verification and (4) Document Management. each module will be implemented following the design and data established during design phase

Throughout the development process the code versions will be regularly reviewed and updated. Researchers will document every iteration by recording changes, issues encountered and the solution that were applied. Each sprint will focus on developing one or more modules, followed by testing and user feedback, as well as reviewing our progress and adjusting backlog priorities based on encountered issues.

3.3.1.3.0 Development and coding practices

During this phase, we will also implement secure coding practices to ensure the system's integrity. And as such, we will add security controls such as input validation and output encoding to all user-provided data to prevent

injection attacks and data corruption.

Multi-factor authentication to verify their identity, make sure the one using the system is the one with the corresponding identity, and based on their identity, least privilege access will be applied to restrict user permissions based on their roles.

Password will not be stored in text, instead hashed and salted storage methods will be implemented, hashing converts the password into a cryptographic value, while salting adds a random value to each password before hashing, using the two would then make hashes unique and resistant to rainbow table attacks. All login events such as logins, failed attempts and unauthorized access will be logged for audit and monitoring. Furthermore, security policies will guide the system's implementation to further strengthen the security, including (1) access control policy, which defines who can access what in the system and under what conditions to prevent unauthorized access to information

(2) password policy, sets the rules for creating, using and managing password to keep accounts secure, to strengthen account security and prevent unauthorized logins

(3) remote access policy, which defines how users can safely connect to the system outside the barangay's local network as well as prevent unauthorized external access and protect the system from remote intrusions.

(4) network security policy, to establish how the system and its

network are protected from external attacks or unauthorized use.

(5) Software and Patch management Policy, ensures that all software components, servers and applications are kept up to date with the latest security patches as attackers will often use exploits on outdated or unpatched methods of breach.

- **3.3.1.5 Testing Phase**

The modules created each sprint will be tested to provide the users with full transparency on the system's status, with great focus on the users to gain additional and substantial feedback, Testing will occur after every sprint and as such the researcher will be conducting numerous testing types in order for the system to be further refined will be conducted during this phase.

There will be two testing phases, one is alpha testing where it will identify and resolve functional, performance and security issues before moving on to the beta testing, which will be tested according to the barangay staff's given requirements

In the alpha test, the researchers will conduct numerous testing such as the (1) **unit testing** to verify that each module of the system works and functions as intended, (2) **integration testing**, to ensure that each module when transferring data in between each module will be consistent, (3) **stress testing** to evaluate the systems stability and performance under a series of heavy loads like simultaneous logins or multiple document uploading, (4) **Compatibility testing** to check whether the system performs consistently

across different browsers, resolutions, aspect ratios and even operating environments that will be used by the barangay. **(5) Reliability testing** to assess the system's ability to operate continuously, and lastly the (6)

Cybersecurity testing

3.3.1.5.0 Cybersecurity testing

Evaluates the robustness of the system's security mechanisms, to ensure protection against potential threats, like an ethical hacking simulation, to identify vulnerabilities in authentication or encryption,

Brute force Testing, on login attempts, testing out password strength and lockout attempts against repeated login attempts, SQL injection testing to verify that all input fields are properly protected against database manipulation, Cross-Site Scripting (XSS) Testing, to ensure that no unvalidated scripts can be executed through user inputs.

Encryption validation which verifies that the encrypted documents cannot be accessed or verified without proper authorization, and aligning the cybersecurity to the NIST Cybersecurity framework to ensure structured assessment and risk mitigation. All identified vulnerabilities and corresponding mitigation actions will be recorded for continuous improvement.

Mobile Application Testing

The mobile authentication application will undergo:

1. Functional Testing – to verify successful scanning, extraction, decryption, and comparison processes.
2. Tamper Detection Testing – to validate detection accuracy when contract files are modified.
3. Performance Testing – to measure scan-to-result response time under normal usage conditions.
4. Compatibility Testing – to ensure consistent operation across supported Android devices and screen sizes.

The detection accuracy rate and authentication response time will be recorded as part of system evaluation metrics.

- **3.3.1.6 Deployment, Review and Launch Phase**

After completing all the Testing, the system will proceed to the deployment, review and launch phase, we will have the most stable build of the system, in which it will be deployed in a secure local or cloud-based environment for pilot testing. The researchers will also demonstrate the system's functionalities, such as document upload, authentication, and verification features. Feedback that will be gathered during this stage will be used to make minor refinements and address any remaining issues.

After deployment, performance monitoring will be implemented to ensure the system's stability and protection against new threats. There will be activities to be carried out such as checking out the system logs to detect

suspicious activity, intrusion attempts or abnormal behavior. And their responses to these incidents regarding threats and swift mitigation. To add to their responses to these incidents, there will be a brief security awareness orientation for end users, as well as simple user guidelines and documentation so they can operate the system safely and confidently without disrupting their regular duties

3.3.2 Implementation

From here will be the technical aspects of the system, the system will be developed using the python programming language under the Django web framework due to its features like Cross-Site Request Forgery (CSRF) protection which ensures that all important actions such as contract uploads or deletions, can only be executed through legitimate and authenticated user requests, SQL injection prevention, and secure password-hashing. As well as django's modular architecture which supports our prototyping during sprints.

For Database management, the system will utilize Firebase Realtime Database as its cloud-hosted NoSQL database as well as its real time data synchronization as well as its dedication authentication for user management, however given the sensitivity of the documents it would be easy to assume that a NoSQL database would be a security risk, it would be if it's not configured properly, by enforcing a strict access control and role based security rules, so that only verified users can access the database, as well as an applied pre-upload encryption and hashing so even if the database were exposed the stored contracts would remain unreadable.

For version control, Github will be used, as Novoseltseva (2024) mentioned, in Github you can track the changes in your code, documentation, as well as make it easy for multiple people to contribute in a project through forks and pull requests. A private repository will be configured so that all system files will remain confidential.

Visual Studio Code will be used for coding and debugging, Figma will be used for interface prototyping and UI/UX design mock-ups, for api testing and cybersecurity vulnerability simulation, Postman will be used.

3.3.3 Cybersecurity Implementation

Multiple layers of cybersecurity mechanisms will be integrated into the system, for Authentication and Access Control it will use Firebase Authentication to regulate access. Role-based access control ensured that only verified barangay officials could upload or modify contract data, while public users will be limited to document verification functions and reporting.

In the database before uploading the PDF contract will be encrypted using the AES-256 encryption so that even if the files are accessed externally, their contents will remain unreadable without the encryption key, upon upload, each contract file generated a unique SHA-256 hash value, this hash would then be embedded into a digital seal image using the Least Significant Bit (LSB) steganographic technique. During verification, the system extracted the hidden hash and compared it with the stored record hash in firebase to determine if the document is authentic. To further strengthen this system, the backend system will be tested using Postman to simulate unauthorized requests and validate Firebase security rules. As well as Ethical Hacking simulations such

as brute-force login attempts, input injection, and cross-site scripting

3.4.0 Population of the Study/Sampling Design

The population of the study includes the barangay officials and employees who participate in the preparation, management, and archiving of government contract documents. This includes the Barangay Captain, Barangay Secretary, Barangay Treasurer, Bookkeeper, and other staff members who directly interact with the system.

The study employed an interpopulation and stratified sampling design. The respondents were grouped based on their roles within the barangay to ensure a balanced representation of perspectives from different functional areas. The strata included an administrative officials who are responsible for approving and validating official records and contracts, Administrative Staff who are responsible for document encoding, storage and release and the Public citizens who will request, receive or verify official barangay documents

Table 3.1 Population Distribution Per Stratum

Stratum	Description	Sample Size
Administrative Officials	Barangay Captain, Secretary, Treasurer, and Bookkeeper responsible for document approval and validation	4

Administrative Staff	Clerks and personnel responsible for encoding, storing, and releasing official records	4
Public Citizens	Residents who request, receive, or verify barangay documents	30
Total		38

$$n = \frac{N}{1 + N(e)^2}$$

Figure 3.6 Sampling Formula Used

N = Sample Size

E = Margin of

Error

3.5.0 Data Gathering Tools and Procedures

The methodology that will be employed for this project will utilize a quantitative research approach centered on requirements elicitation through an in-depth interaction with a key administrative official. As well as The primary goal was to obtain comprehensive functional and non-functional requirements for the proposed system

- **Interviews and Observation**

The principal instrument for data collection was a Semi-Structured Interview Protocol. Which was designed to ensure both standardization and flexibility during the requirements gathering phase. Before the Interview, the researchers also prepared a list of

core questions that was organized into three areas: Current Workflow Analysis, which focuses on the existing manual processes for document handling; Problem Identification, aimed at uncovering security vulnerabilities and potential risks, and System requirements elicitation to find out the desired features and necessary constraints for the digital system.

- **System Testing Phases**

To ensure the system works as intended before deployment, Testing phases will be conducted, such as the (1) Alpha testing which will be carried out internally, it will simply be identifying bugs and resolving them, validating that all the aforementioned modules function correctly, and penetration testings to check the system's security after we have completed a series of tests, we will be able to move on and do Beta Testing, in which end users will participate and evaluate its usability and performance. During this phase, data will be gathered through the **User Acceptance Testing (UAT)** and the **System Accuracy Testing**

- **System Accuracy Testing**

The procedure was conducted to determine the reliability and precision of the system in processing, encrypting and storing data. A series of predefined test cases were executed to compare the system's actual outputs against the expected results.

Furthermore we will also test the system's reliability and responsiveness through how quickly the system identifies the faults, and how quickly it could resolve

faults or security-related incidents during testing. During this phase, intentional faults and simulated anomalies such as invalid inputs, unauthorized access attempts or failed encryption tasks will be introduced to evaluate how promptly the system detects and responds to such events.

To ensure the robustness and reliability of the system’s encryption and verification process, a series of test cases will be executed, Each test case will check how accurately the system encrypts decrypts and verifies document authenticity under different content structures, each varying on text length formatting and character composition.

Test Case ID	Document Type	Description	Expected Output	Purpose
TC-01	Single Word	A text file containing only one word (e.g., "Barangay")	Hash and decrypted file match original	Validate minimal text handling
TC-02	Short Phrase	Two to three words (e.g., "Barangay Contract Form")	Hash and decrypted file match original	Test short document encryption consistency
TC-03	Paragraph Document	Four paragraphs of text without special characters	Hash and decrypted file match original	Evaluate encryption/decryption over longer text blocks
TC-04	Special Character File	Text containing colons, semicolons, commas, and symbols (e.g., "Contract: RA 9184; Section 53, P500,000.00")	Hash and decrypted file match original	Assess behavior when encoding special symbols
TC-05	Mixed Content Document	Document with both text and embedded image or scanned signature	Combined hash (text + image) matches original	Test combined hash integrity of text and image data
TC-06	Re-saved / Altered Document	Re-saved version of original PDF with minor textual edits	System flags as "Possibly Tampered"	Validate system's ability to detect even minor changes

Table 3.2 Document Test Cases

- **User Acceptance Testing (UAT)**

This is where researchers evaluate the system functionality. Participants of this test will check if login, document management, uploading or authentication works, based on accuracy, speed, and reliability. Researchers will then use the results from the series of tests to validate whether the system works normally under conditions. Researchers will be using performance metrics using logging mechanisms such as Time on Task which is how long it would take to complete a certain function, and its success/error rate to check the efficiency and performance in securing and verifying these PDF files.

3.5.1 Procedures

The data gathering process followed a strict procedure to ensure ethical conduct and methodological integrity.

1. **Preparation and Authorization:** Formal protocol was initiated by drafting and submitting a letter of request to the barangay captain of Sto. Niño, Biñan City, Laguna. This letter detailed the purpose, scope of the research and the nature of the interview. Once authorization was secured, the schedule and venue were finalized as well as the interviewee's consent for a recording in order to create a transcript.
2. **Interview Questions Refinement:** Before interview, the questions underwent an internal review by the research group so the questions are aligned with the functional goals of the system and the practical constraints of the barangay

setting. Thus conducting the interview.

3. **Quantitative Data Collection:** Once the prototype is developed, User Acceptance Testing will be conducted to the barangay staff. Participants will be asked to perform predefined system tasks while their interactions, task completion time, and error occurrences were recorded.
4. **Data Transcription:** Interview recordings will be transcribed. And then the transcripts will be analyzed, translating qualitative data into system requirements. Involving rigorous review of the data to identify recurring concepts, and identifying the stated needs such as the reliance on physical copies and the need for training. These identified themes were systematically mapped for what the system must do and how the system should perform.
5. **Data analysis:** The results of the System Accuracy Testing were presented as percentages to indicate the system's precision, while UAT results will be analyzed using weighted mean and standard deviation to evaluate user satisfaction and acceptance.

All participants were informed of the study's purpose and gave their consent prior to participation.

3.6.0 Data Analysis Plan

This study involved the barangay captain of Sto. Niño, Biñan laguna, as well as it's barangay officials and staff. Based on the data that will be gathered. Since mentioned in the research design that we will be using quantitative, we will be analyzing quantitative data gathered from the system's performance and user evaluation. The main analyses consist of the

System Accuracy Testing and User Acceptance Testing (UAT), which quantitatively assess the performance of the system. Additionally, interviews will be conducted to gather supporting insights from barangay personnel regarding their experiences, The interview responses will be used to support and contextualize the quantitative findings.

3.6.1 Interview and Observation

Qualitative data will be primarily obtained through interviews and observations conducted in barangay Sto. Nino, Aimed at capturing the lived experiences, perceptions and the Processes they do regarding, document management, procurement processes, and security, as well as human-centered insights and offer contextual depth and holistic perspectives. The data analysis will follow the process of Thematic Analysis as proposed by Braun and Clarke (2006). The researchers will perform analysis of data such as transcribing the audio recordings of the interviews in order to gain a comprehensive understanding of the participants' responses. Each relevant statement or response from participants will be assigned a code or label that represents a specific concept like document management, manual handling, in order to make the data more easy to understand, and then group these even more into themes that capture significant insights. And then reviewing these said themes against the data to ensure coherence, repetitive themes will be merged and themes that lack data will be used to gather more data. After all is said and done, we will then interpret the final themes to study's objectives, linking participants' insights to the system's design features.

3.6.2 System Accuracy Testing

This will focus on evaluating the system in processing, encrypting and decrypting records, accuracy will be determined by comparing the number of correctly decrypted and

verified outputs against the total number of encryption, which aims to ensure the system performs its intended operations in authentication, file handling and encryption.

The process involved subjecting the system to a series of test cases which are designed to represent common operations within the barangay's workflow for each test case, the expected output was compared with the actual output generated by the system, any deviation from the expected result was recorded as an error, while correct outputs were counted as successful operation, accuracy was computed using the following figure:

$$\text{Accuracy (\%)} = \frac{\text{Num of correct outputs}}{\text{Total Test Cases}} \times 100$$

Figure 3.7 Accuracy Formula

This formula quantifies how many operations the system performed correctly out of the total test, a higher percentage would indicate that the system is reliable.

In addition to this, the accuracy of the system's encryption and decryption functions will be specifically evaluated, in which researchers will be performing various encryption and decryption on various document files to verify whether the decrypted output matched the original file. Each successful match will indicate a correct operation, while discrepancy will be logged as an error. The researchers will use the hash values of both original and encrypted files and check if both hash outputs are identical or not.

$$\text{Integrity (\%)} = \frac{\text{Num of matching outputs}}{\text{Total Test Cases}} \times 100$$

Figure 3.8 Integrity Formula

The combination of correct outputs and integrity accuracy by matching hash provides a complete quantitative measure of the system's data reliability and security performance.

Results of the system accuracy testing would then be summarized, indicating each test case the expected result

Table 3.3 interpretation of System Accuracy Testing

Point	Accuracy Range	Interpretation
1	95-100%	Highly reliable
2	90-94%	reliable
3	80-89%	Acceptable
4	70-79%	Noticeable errors
5	Below 70%	Unreliable

Both System Accuracy and Integrity Accuracy will be using the same table, but both will be interpreted separately.

3.6.3 User Acceptance Test

Questionnaires will be given to the respondents to evaluate the prototype and assist in the enhancement of the quality of the device and applications. The evaluation of the results will be assessed via calculating the mean or average utilized to modify the prototype further as shown by figure 3.5

$$\bar{x} = \frac{\sum (w \cdot x)}{\sum w}$$

Figure 3.9 Weighted Mean Formula

Feedback received during the User Acceptance Testing (UAT) phase will be gathered, organized into distinct categories encompassing design, functionality and reliability concerns. Evaluation of user suggestions will consider their potential impact and feasibility, with the identification of recurring themes within the feedback to inform decision-making in the software development process. To interpret the results, the following scale will be used:

Table 3.4 Interpretation of the UAT Scale

Point	Accuracy Range	Interpretation
1	1.0-1.80	Strongly Disagree
2	1.81-2.60	Disagree
3	2.61-3.40	Neutral/Uncertain
4	3.41-4.20	Agree
5	4.21-5.00	Strongly Agree

3.6.5 Integration

After performing all data analyses, we will integrate results from the data gathering tools to provide a comprehensive assessment of the proposed system. The System Accuracy Test will establish the system's technicality, determining whether the software correctly processes and secures data through the encryption process, and making sure the documents maintain integrity, the User Acceptance Testing will then reflect on the user's perceived usability, and satisfaction during the system interaction.

Integration of these findings will follow a triangulated interpretation approach, in which the integrations process follows the approach of Creswell (2018) who emphasized combining quantitative indicators to form a comprehensive interpretation of system effectiveness such as the quantitative technical metrics like accuracy and integrity alongside the UAT. Additionally, insights gathered from interviews will serve as supporting context to explain observed trends.

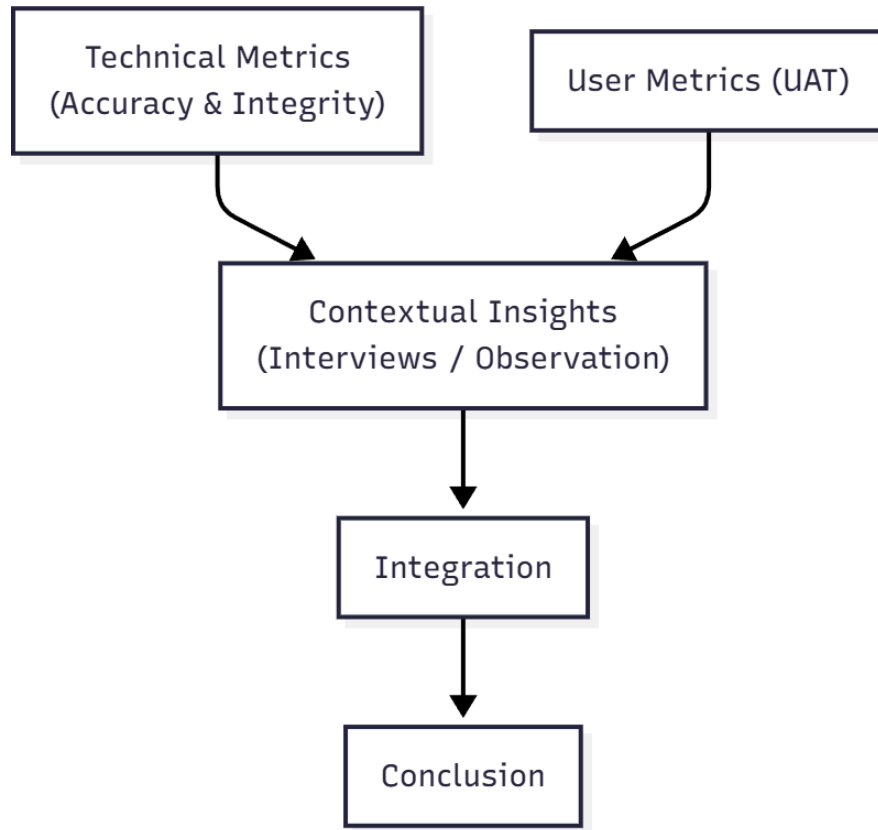


Figure 3.10 Triangulated Interpretation Diagram

Figure shown in 3.6 is the triangulated interpretation diagram that will be used to interpret the data, The Technical Metrics represents all the system-level performance tests, proof that the system works correctly and secure, which includes accuracy and integrity of the system, while the user metrics would then be the user side evaluation, how the users perceive the system and its usability. Contextual has the reasons behind the numbers, the user attitudes, workflows, human factors, the integration factor is the core of the triangulation that simply compares and all the evidence is synthesized, it checks for convergence which is where results are in agreement (system is accurate, users are satisfied), Divergence (secure system but poor usability) and

complementary which explains the low scores via interview. And lastly the output which is where researchers will draw practical conclusions based on the integrated evidence.

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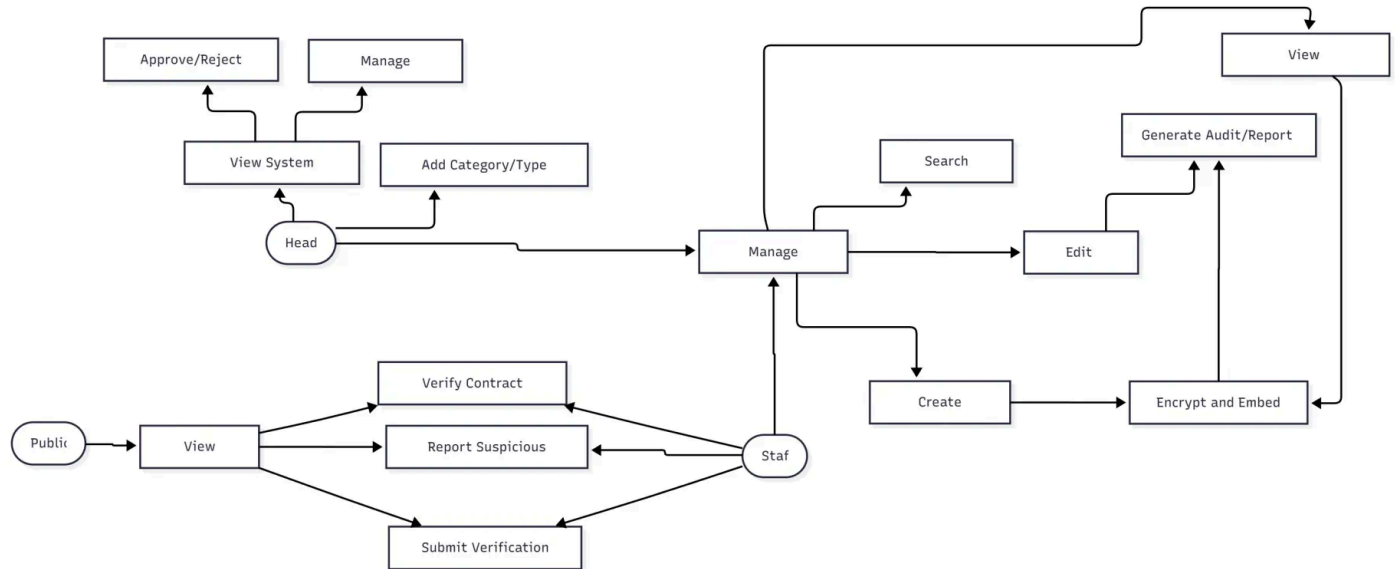
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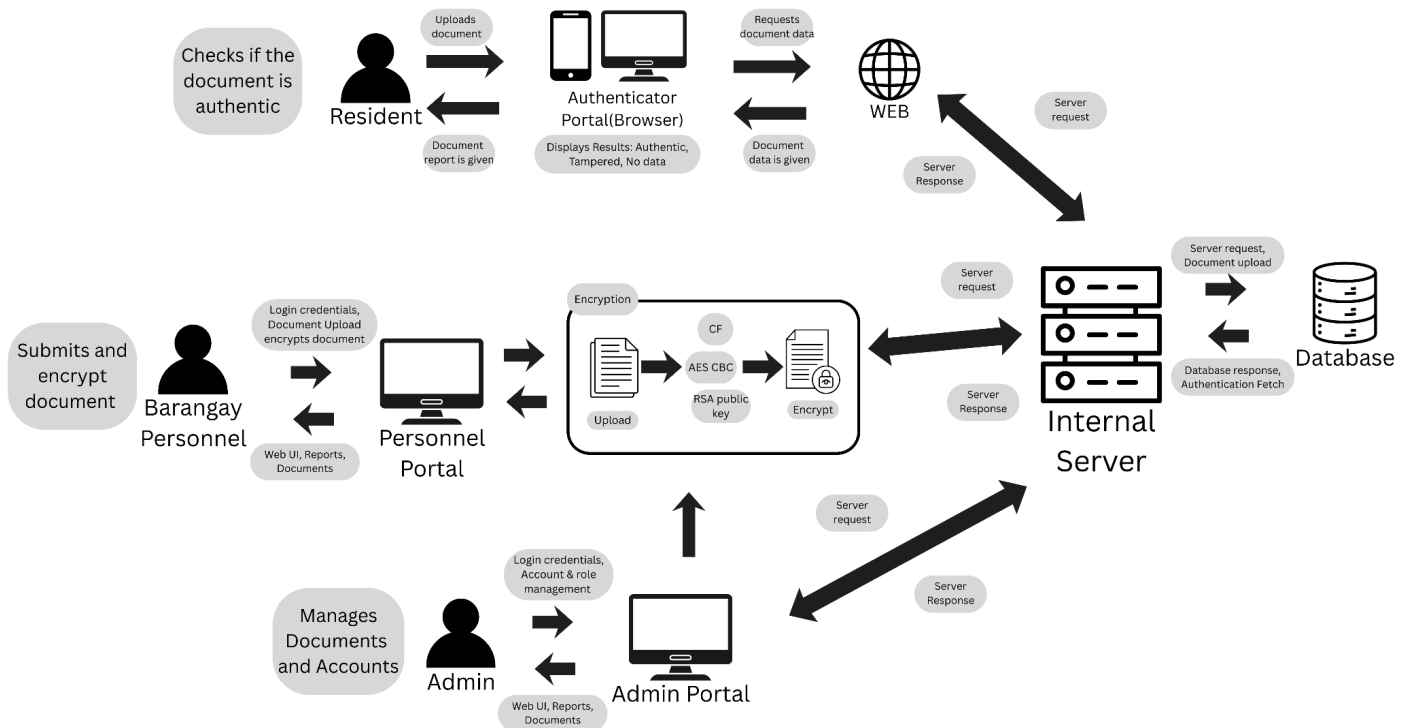
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Appendices

Appendix A. Data Flow Diagram

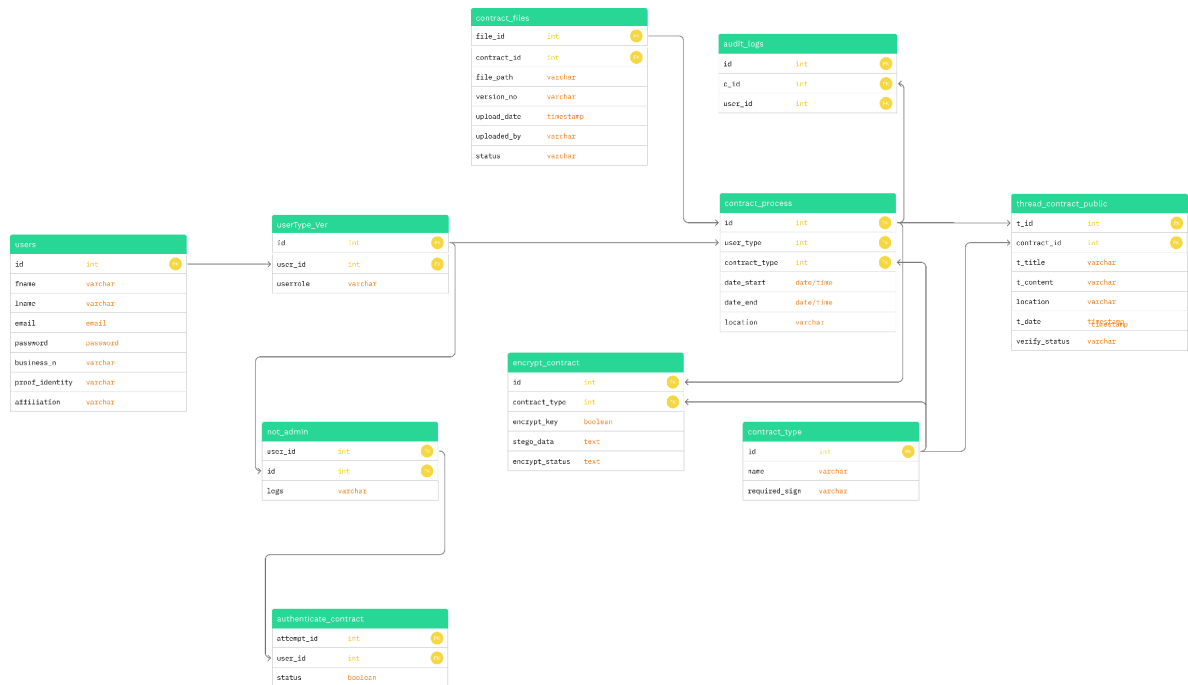


Appendix B. System Architecture Diagram



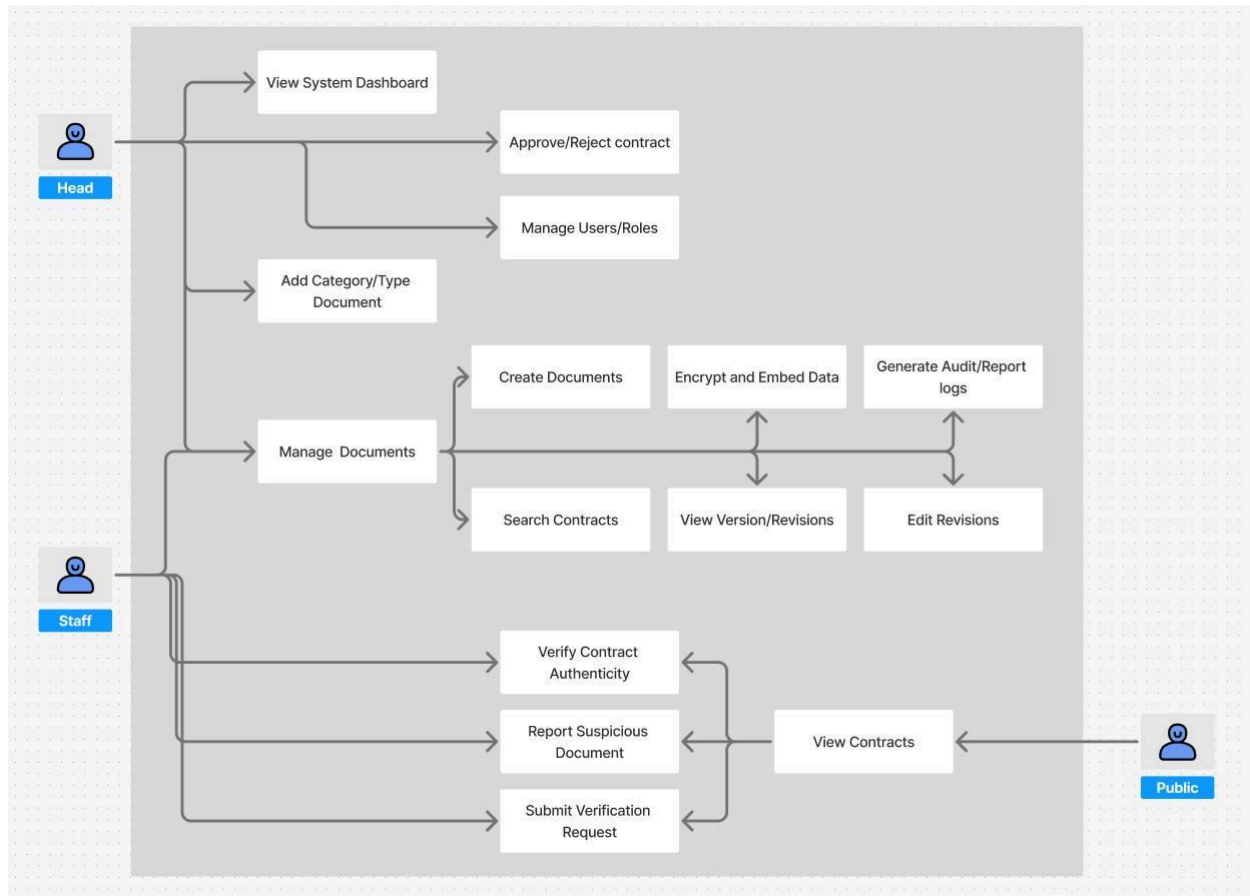
The figure illustrates the overall system architecture of the proposed system, highlighting the flow of data and interactions among users

Appendix C. Entity Relationship Diagram



The ERD illustrates how user data, contract processing, encryption, and verification modules interact to ensure secure document management and public transparency.

Appendix D. Use Case Diagram



This use case diagram illustrates the overall interaction between the users and the system this shows how each Contracts This use case diagram illustrates the overall interaction between the users and the system

Appendix E. Sprints Deliverables

Sprint	Duration	Main Deliverable	Features
Sprint 1	Weeks 1–2	User Management	Core user registration, login/logout, role-based permissions, and audit logging features.
Sprint 2	Weeks 3–4	Encryption & Authenticator	Implementation of encryption algorithms for documents, authentication logic, and decryption features for authorized users.
Sprint 3	Weeks 5–6	Document Management	Document upload/retrieval system, version control, metadata tagging, and access tracking functionalities.
Sprint 4	Weeks 7–8	Report Dashboard	Data analytics dashboard with charts, activity summaries, exportable reports, and responsive visualization tools.
Sprint 5	Weeks 9–10	Integration Testing	Integration of all modules, debugging, and system-wide testing for interoperability and performance.
Sprint 6	Weeks 11–12	Final Review & Tweaks	System demonstration, refinement based on feedback, final documentation, and deployment preparation.

Appendix F. Interview Questions:

General Procurement Practices

What types of contracts does your barangay usually engage in?

On average, how many procurement contracts are processed in a month?

Does your barangay use any information system or digital platform to manage procurement

activities? If yes, can you describe how it works?

Procurement Process and Approval

Can you describe the step-by-step process your barangay follows for drafting, reviewing, and approving procurement contracts? At what level or office are contracts typically approved?

How are contracts evaluated and approved in accordance with Republic Act 9184?

After a contract is signed and approved, which office or staff member is responsible for reporting or monitoring the implementation?

RA 9184 Implementation and Compliance How familiar is your barangay with the guidelines and procedures under Republic Act 9184?

What challenges does your barangay face in complying with the requirements of RA 9184?

Has the barangay received any training or orientation regarding the procurement rules

under RA 9184?

In your opinion, how has RA 9184 affected transparency and efficiency in your barangay's procurement process?

How are procurement documents currently stored — digitally or in hard copy?

Are there any existing procedures to verify whether a document is genuine or has been tampered with?

Has your barangay experienced issues such as lost, altered, or falsified documents?

Digitalization and Security

Has the barangay considered adopting digital tools or systems to manage procurement records?

What concerns do you have about using digital systems for handling sensitive government contracts (e.g., privacy, tampering, technical difficulties)?

Appendix G. Interview Questions SIGNED

Interview Questions:

General Procurement Practices

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In your opinion, how has RA 9184 affected transparency and efficiency in your barangay's procurement process?

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Are there any existing procedures to verify whether a document is genuine or has been tampered with?

Has your barangay experienced issues such as lost, altered, or falsified documents?

Digitalization and Security

Has the barangay considered adopting digital tools or systems to manage procurement records?

What concerns do you have about using digital systems for handling sensitive government contracts (e.g., privacy, tampering, technical difficulties)?


WILFREDO DENNIS M. HERNANDEZ

Appendix H. UAT Questions

No	Statement	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
		5	4	3	2	1
User Interface Design						
1	The website is user-friendly and visually appealing.					
2	I was able to locate the information I needed easily.					
3	The website layout and navigation are clear and intuitive.					
Functionality						
4	Uploading/viewing documents is simple.					
5	I can easily review the number of pending incidents and the type of report being sent					

6	The system shows if a document is real					
Usability						
6	I did not experience any technical difficulties while using the website (e.g., slow loading times, broken links).					
Reliability						
7	The website functions without frequent crashes or errors.					
8	The system keeps documents safe from being changed or stolen.					
9	The documents integrity and contents are still the same after encryption					

Appendix I. Letter of Request of Interview.

October 9, 2025

Hon. Dennis Hernandez
Barangay Captain
Barangay Sto. Niño, City of Biñan

Greetings of Peace and Joy!

I hope this letter finds you in good health and high spirits. We are BSIT students taking up IT200-1P (Capstone 1) of **Mapúa Malayan Colleges Laguna**.

We are writing to request an interview with you as part of a beneficiary of the Capstone Project entitled **"A web-based Government Information System with Authenticity Verification using steganography-based encryption technique"**.

The project aims to provide an alternative solution to evaluate the integrity of the government contract in support of the Republic Act 9184 which pertain into **"An Act Providing for Modernization, Standardization and Regulation of the Procurement Activities of the Government for other Purposes."** Through the interview, we will gain a deeper understanding of the procurement contract of the barangay; your insights and experiences are invaluable to our research. The interview will be conducted at your convenience, and we are willing to adjust to your schedule. We anticipate that the interview will take a few minutes, please rest assured that all information shared during the interview will be treated with the utmost confidentiality, and your privacy will be respected.

Feel free to let us know, and we will accommodate your requests to the best of our ability. Your participation in this research would contribute significantly to our understanding of the process of the procurement contracts you handle, and it would be greatly appreciated. Thank you for considering our request. We look forward to the opportunity to learn from your experiences and insights.

You can reach us through 09297942550 or e-mail us at griffindeleon@gmail.com.

Sincerely,

Charles Go
Dustin Domingo
Griffin De Leon

Noted by:



Adomar L. Ilao

BSIT Program Chair
Mapúa Malayan Colleges Laguna
alilao@mcl.edu.ph (09228197409)

Appendix J. Interview Letter Signed

October 9, 2025

Hon. Dennis Hernandez
Barangay Captain
Barangay Sto. Niño, City of Biñan

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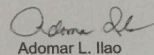
Feel free to let us know, and we will accommodate your requests to the best of our ability. Your participation in this research would contribute significantly to our understanding of the process of the procurement contracts you handle, and it would be greatly appreciated. Thank you for considering our request. We look forward to the opportunity to learn from your experiences and insights.

You can reach us through 09297942550 or e-mail us at griffindeleon@gmail.com.

Sincerely,

Charles Go
Dustin Domingo
Griffin De Leon

Noted by:

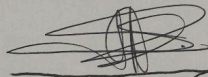


Adomar L. Ilao

BSIT Program Chair

Mapúa Malayan Colleges Laguna

alilao@mcl.edu.ph (09228197409)


WILFREDO DENNIS M. HERNANDEZ

Appendix K. INTERVIEW TRANSCRIPT:

Griffin: Hello ako si Griffin Gil De Leon at ang kasama ko ngayon ay ang barangay captain ng sto nino binan, sir Wilfredo Hernandez

Barangay Captain: Hello, ako si Wilfredo Hernandez, Captain ng Sto nino binan for 2 years

Griffin: firstly ano po yung usually ine-engage ng barangay na mga contracts or documents?
(What types of contracts does your barangay usually engage in?)

Barangay Captain: Mga maliliit na kontrata, kasi kalimitan naman dito sa barangay, kapag malaki na project, city ang sponsor, katulad ng pinapagawang school, na galing kay senator joel villanueva, so through Cong. Len dahil sa kanila ni request ko sa kanila, galing siya sa national pero kung ang paguusapan ay yung mga binibili, na mga gamit o yung mga ginagawa ng barangay, maliit lang naman kasi ang budget natin, eh napakaliit din kasi maliit lang na barnaggay tayo, next question

Griffin: Next question po ay, ano po yung mga na pprocesso niyo na kontrata?

Barangay captain: normally up to 1-2 times lang eh, kunyari yung cctv, eh mahilig 300,000php ang halaga so kailangan siyang iba bidding, kailangan siyang may contractor, kailangan siya meron na bidding na mangyayari

Griffin: Sa mga documents niyo po gumagamit po ba kayo ng digital o halos physical lang?

Barangay Captain: puro physical kami

Griffin: magtatanong po ako sa proceso niyo po sa procurement pano po yung step by step niyo dun, pano po yung pagddraft niyo, pano po yung pag review, or yung pag approval. Kung pwede po itanong.

Barangay Captain: Ganito yun, halimbawa may project kami dito, so through resolution during budget, katulad ngayon naggagawa na kami ng budget, so sa budget inaalocate na namin yung mga project gusto na namin gawin o kung may mga gamit na kailangan sa barangay na gusto namin bilhin pag ang halaga, pag ang halaga ay 99,000 pababa, hindi siya kailangan ng bidding, pwede na namin kaagad mabili, pero kapag ang halaga ay more than 99,000 so let's say halimbawa 102k, 200k, 800k, 1 million kung bibili ng firetruck, two million, so pag ka ganun ibibidding mo na siya, kukuha ka ng tatlong supplier, ibibidding mo na. Ipag-ugnayan ang barangay meron pag-uusap, nagkakaroon ng proper documentation at ng resolution na may nagmumungkahi may pumapangalawa at sinasangayon ng buong sanggunian na bibili ng ganun gamit, and then nakikipag ugnayan kami sa accounting sa COA (Commission on Audit) at icheck ni treasurer yung budget kung may available na budget, tapos yung supplier, maghahanap ka rin ng supplier na mas mura, saka kami mag pproceed.

Griffin: kapag aabot ng isang million kagaya nung firetruck na sinabi niyo, aabot pa po ba ito sa mayor or kayo na ang naghahandle?

Barangay Captain: Kami lang, pag kaya ng barangay kami lang.

Griffin: ano po yung alam niyo sa Republic Act 9184?

Barangay Captain: 9180?

Griffin: 9184 po

Barangay Captain: ay hindi ako familiar

Griffin: parang lang po siya maging transparent yung paghahandle niyo ng documents, yung pera ay magamit ng maayos, para walang favoritism sa supplier at yung processo ay makita ng publiko. And susunod ko na tanong ay ano po ang mga challenges niyo sa pagsunod dito sa republic act na ito?

Barangay Captain: Siguro yung kakulangan ng training. Since narinig ko lang naman to sa seminar at hindi naman lahat ng staff dito kabisado yung mga requirements ng RA 9184, minsan din mahirap sundan lahat ng proseso lalo na kung maliit lang pondo at kulang kulang ang tauhan, kaya umaasa kami sa COA kung may tanong.

Griffin: paano niyo po stino-store yung mga procurement documents ninyo?

Barangay Captain: Physical copies lahat. Naka-file yan sa opisina tapos ilalagay sa lock cabinet, yung Bookkeeper namin ang in charge diyan.

Griffin: May procedure po ba kayo para ma-verify kung totoo and documents o kung may nagbago o natamper sa mga documents?

Barangay Captain: Oo, kasi lahat ng documents may pirma at stamp ng barangay, may logbook din kung kailan lumabas o ginamit ang document, kaya madaling ma-trace.

Griffin: naka experience na po ba kayo ng mga nawalang o binagong documents?

Barangay Captain: wala pa naman, kasi maingat talaga kami sa record-keeping, pero syempre mangyayari pa rin yan, kapag matagal na yung papel, nagfefade o kaya nasisira.

Griffin: na consider niyo na po bang gumamit ng digital tools para sa mga documents na ginagamit ninyo?

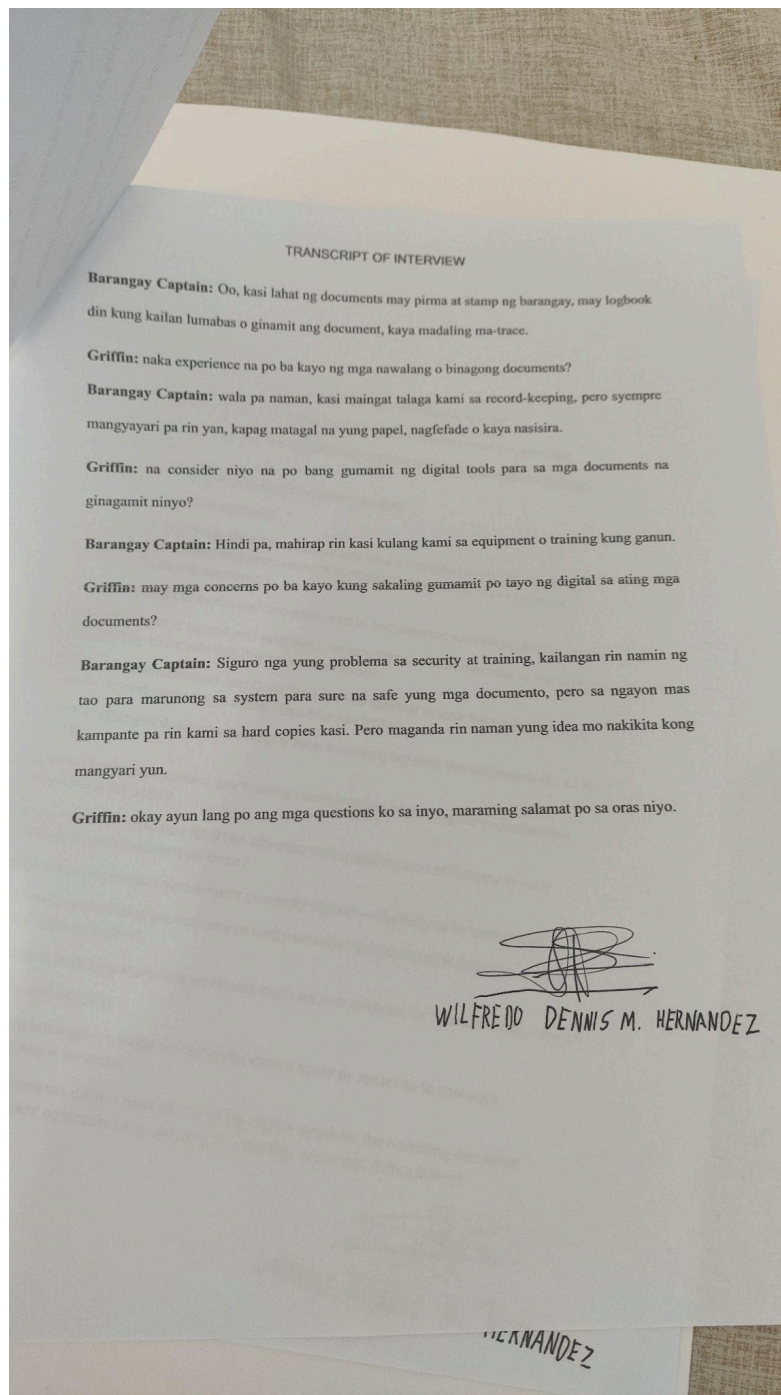
Barangay Captain: Hindi pa, mahirap rin kasi kulang kami sa equipment o training kung ganun.

Griffin: may mga concerns po ba kayo kung sakaling gumamit po tayo ng digital sa ating mga documents?

Barangay Captain: Siguro nga yung problema sa security at training, kailangan rin namin ng tao para marunong sa system para sure na safe yung mga documento, pero sa ngayon mas kampante pa rin kami sa hard copies kasi. Pero maganda rin naman yung idea mo nakikita kong mangyari yun.

Griffin: okay ayun lang po ang mga questions ko sa inyo, maraming salamat po sa oras niyo.

Appendix L. INTERVIEW TRANSCRIPT SIGNED:



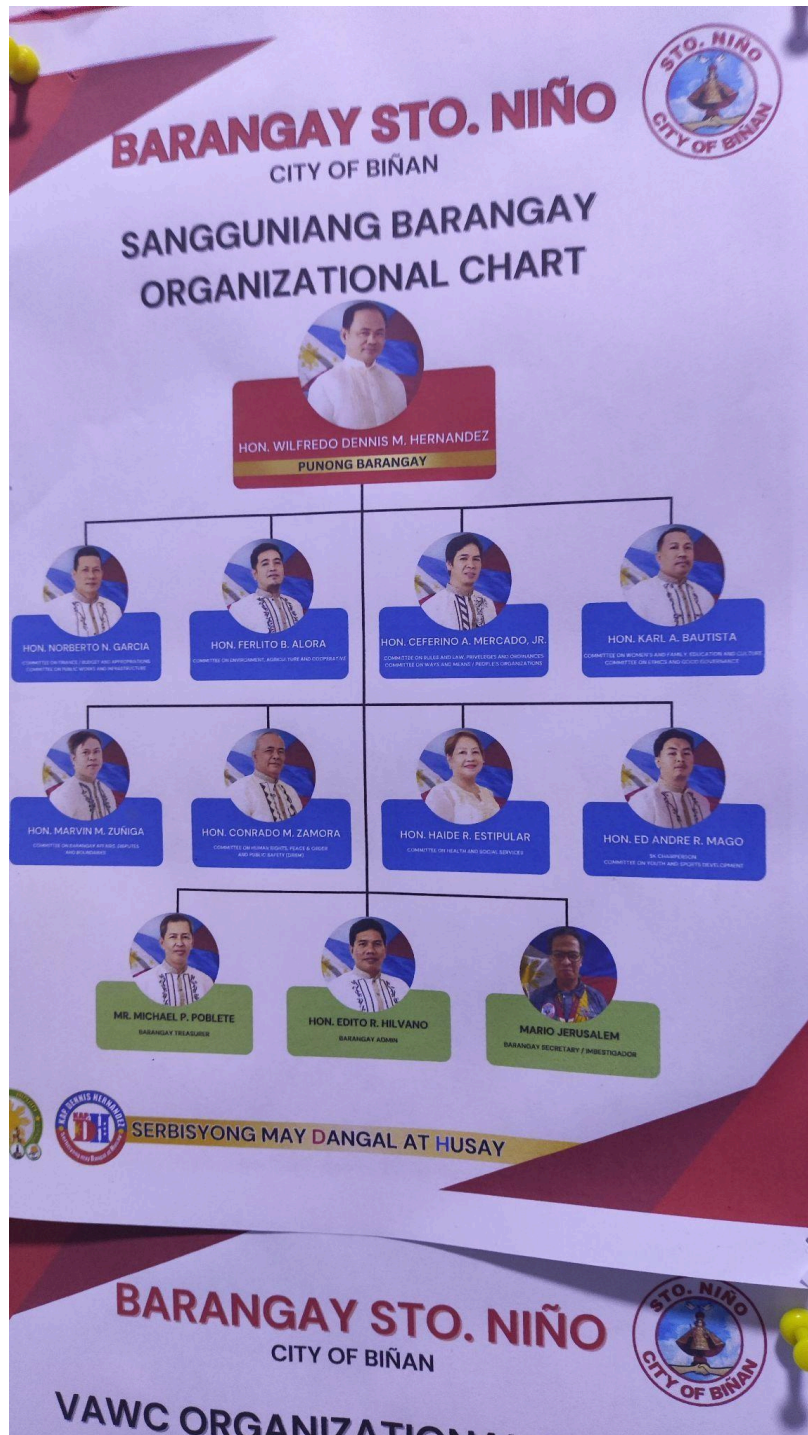
Appendix M. PROOF OF INTERVIEW:



Appendix N. Location of Barangay



Appendix O. Organizational Chart



Appendix P. Annual Procurement Plan (Barangay Sto. Niño, 2025)

Annual Procurement Plan

Republic of the Philippines
Province of Laguna
City of Iloilo

ANNUAL PROCUREMENT PLAN FOR THE YEAR 2025

Name of Barangay: STO. NIÑO
Program Control No.:
Department/Office:

No.	Description	Unit Cost	Qty	Unit	PLANNED AMOUNT		Contingency	Total	Distribution				
					Regular	Contingency			1 st Quarter	2 nd Quarter	3 rd Quarter	4 th Quarter	
1	Electricity				200,000.00		50,000.00		50,000.00		50,000.00		50,000.00
2	Water Expenses				18,000.00		4,500.00		4,500.00		4,500.00		4,500.00
3	Repair & Maintenance (barangay vehicle)				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
4	Removal/Rebuild fence for barangay				250,000.00		100,000.00		100,000.00		100,000.00		100,000.00
5	Traveling Expenses				150,000.00		100,000.00		100,000.00		100,000.00		100,000.00
6	Training Expenses				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
7	Transportation				40,000.00		10,000.00		10,000.00		10,000.00		10,000.00
8	Supplies and Materials (Office Supplies)				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
9	Other supplies and materials expenses				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
10	Membership (personal dues)				200,000.00		200,000.00		200,000.00		200,000.00		200,000.00
11	Public Works				10,000.00		10,000.00		10,000.00		10,000.00		10,000.00
12	Medical Assistance				60,000.00		10,000.00		10,000.00		10,000.00		10,000.00
13	Donations				100,000.00		100,000.00		100,000.00		100,000.00		100,000.00
14	Drugs and Medicines				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
15	Health and Nutrition				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
16	Barangay Assembly				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
17	Cultural Activities				100,000.00		50,000.00		50,000.00		50,000.00		50,000.00
18	Other Maintenance & Operating Expenses				400,000.00		100,000.00		100,000.00		100,000.00		100,000.00
					2,000,000.00		1,100,000.00		3,100,000.00		3,100,000.00		3,100,000.00

Prepared by: 
MICHAEL P. P. P.
Barangay Treasurer

Approved by: 
WILFREDO DENIS HERNANDEZ
Punong Barangay

COMPLIANT
2nd & 3rd QUARTER 2024
AND ANNUAL DOCUMENTS
BARANGAY FULL DISCLOSURE POLICY
BOARD MONITORING

Appendix Q. Disbursement Records

Republic of the Philippines
City of Biñan
Barangay STO. NINO

ITEMIZED MONTHLY COLLECTION AND DISBURSEMENT For the Month of JUNE 2025

DATE	COLLECTION		DATE	DISBURSEMENT	
	PARTICULARS	AMOUNT		PARTICULARS	AMOUNT
1	NTA	894,018.00	2	janitorial materials	49,760.00
15	RPT	117,574.82	2	peace and order uniform	44,675.00
15	CTC/BUS PERMIT/CLEARAN	8,300.00	2	gasoline	29,748.66
			10	repair and maintenance of patrol	50,455.00
			10	construction materials (fencing gate of	55,874.00
			10	disinfectant solutions	52,156.00
			10	signage materials	50,435.00
			10	office supplies	50,299.00
			10	drugs and medicines	50,230.00
			10	Telephone	1,299.00
			10	electricity -1985656161	24,239.36
			17	water utilities 30738845	958.21
			17	metal filing cabinet	49,876.00
			17	rice	49,780.00
			23	electricity -1256033706	940.75
			23	electricity -2103837092	3,409.22
			23	water utilities 3176414	264.77
			23	medical assistance	5,000.00
			23	medical assistance	5,000.00
			23	cleaning and declogging	49,000.00
			27	livelihood prog ram rug making	49,840.00
			27	payroll	430,260.00
			27	pag ibig	600.00
TOTAL COLLECTION		1,019,892.82	TOTAL EXPENSES		1,104,296.97

Prepared by:

MICHAEL P. POBLETE
Barangay Treasurer

Noted by:

WILFREDO DENNIS M. HERNANDEZ
Barangay Chairman

Appendix R. Republic Act No. 9184 – Government Procurement Reform Act (Sections: 3,4,8,10, 12, 35)

SEC . 3. Governing Principles on Government Procurement. – All

procurement of the national government, its departments, bureaus, offices and agencies, including state universities and colleges, government-owned and/or -controlled corporations, government financial institutions and local government units, shall, in all cases, be governed by these principles:

- (a) Transparency in the procurement process and in the implementation of procurement contracts.
- (b) Competitiveness by extending equal opportunity to enable private contracting parties who are eligible and qualified to participate in public bidding.
- (c) Streamlined procurement process that will uniformly apply to all government procurement. The procurement process shall be simple and made adaptable to advances in modern technology in order to ensure an effective and efficient method.
- (d) System of accountability where both the public officials directly or indirectly involved in the procurement process as well as in the implementation of procurement contracts and the private parties that deal with government are, when warranted by circumstances, investigated and held liable for their actions relative thereto.

(e) Public monitoring of the procurement process and the implementation of awarded contracts with the end in view of guaranteeing that these contracts are awarded pursuant to the provisions of this Act and its implementing rules and regulations, and that all these contracts are performed strictly according to specifications.

SEC . 4. Scope and Application. – This Act shall apply to the Procurement of Infrastructure Projects, Goods, and Consulting Services, regardless of source of funds, whether local or foreign, by all branches and instrumentalities of government, its departments, offices and agencies, including government-owned and/or -controlled corporations and local government units, subject to the provisions of Commonwealth Act No.

138. Any treaty or international or executive agreement affecting the subject matter of this Act to which the Philippine government is a signatory shall be observed.

SEC. 8. Procurement By Electronic Means. – To promote transparency and efficiency, information and communications technology shall be utilized in the conduct of procurement procedures. Accordingly, there shall be a single portal that shall serve as the primary source of information on all government procurement. The G-EPS shall serve as the primary and definitive source of information on government procurement. Further, the GPPB is authorized to approve changes in the procurement process to adapt to improvements in modern technology, provided that such modifications are consistent with the provisions of Section 3 of this Act.

To take advantage of the significant built-in efficiencies of the G-EPS and the volume discounts inherent in bulk purchasing, all Procuring Entities shall utilize the G-EPS for the procurement of

common supplies in accordance with the rules and procedures to be established by the GPPB. With regard to the procurement of non-common use items, infrastructure projects and consulting services, agencies may hire service providers to undertake their electronic procurement provided these service providers meet the minimum requirements set by the GPPB.

SEC . 10. Competitive Bidding. – All Procurement shall be done through

Competitive Bidding, except as provided for in Article XVI of this Act.

SEC . 12. Functions of the BAC. – The BAC shall have the following functions:

advertise and/or post the invitation to bid, conduct pre-procurement and pre-bid

conferences, determine the eligibility of prospective bidders, receive bids, conduct the

evaluation of bids, undertake post-qualification proceedings, recommend award of contracts

to the Head of the Procuring Entity or his duly authorized representative: Provided, That in the event the Head of the Procuring Entity shall disapprove such recommendation, such disapproval shall be based only on valid, reasonable and justifiable grounds to be expressed

in writing, copy furnished the BAC; recommend the imposition of sanctions in accordance

with Article XXIII, and perform such other related functions as may be necessary, including

the creation of a Technical Working Group from a pool of technical, financial and/or legal

experts to assist in the procurement process.

SEC . 35. Failure of Bidding. – There shall be a failure of bidding if:

(a) No bids are received;

(b) No bid qualifies as the Lowest Calculated Responsive Bid or Highest Rated Responsive Bid; or,

(c) Whenever the bidder with the highest rated/lowest calculated responsive bid refuses, without justifiable cause to accept the award of contract, as the case may be.

Under any of the above instances, the contract shall be re-advertised and re-bid.

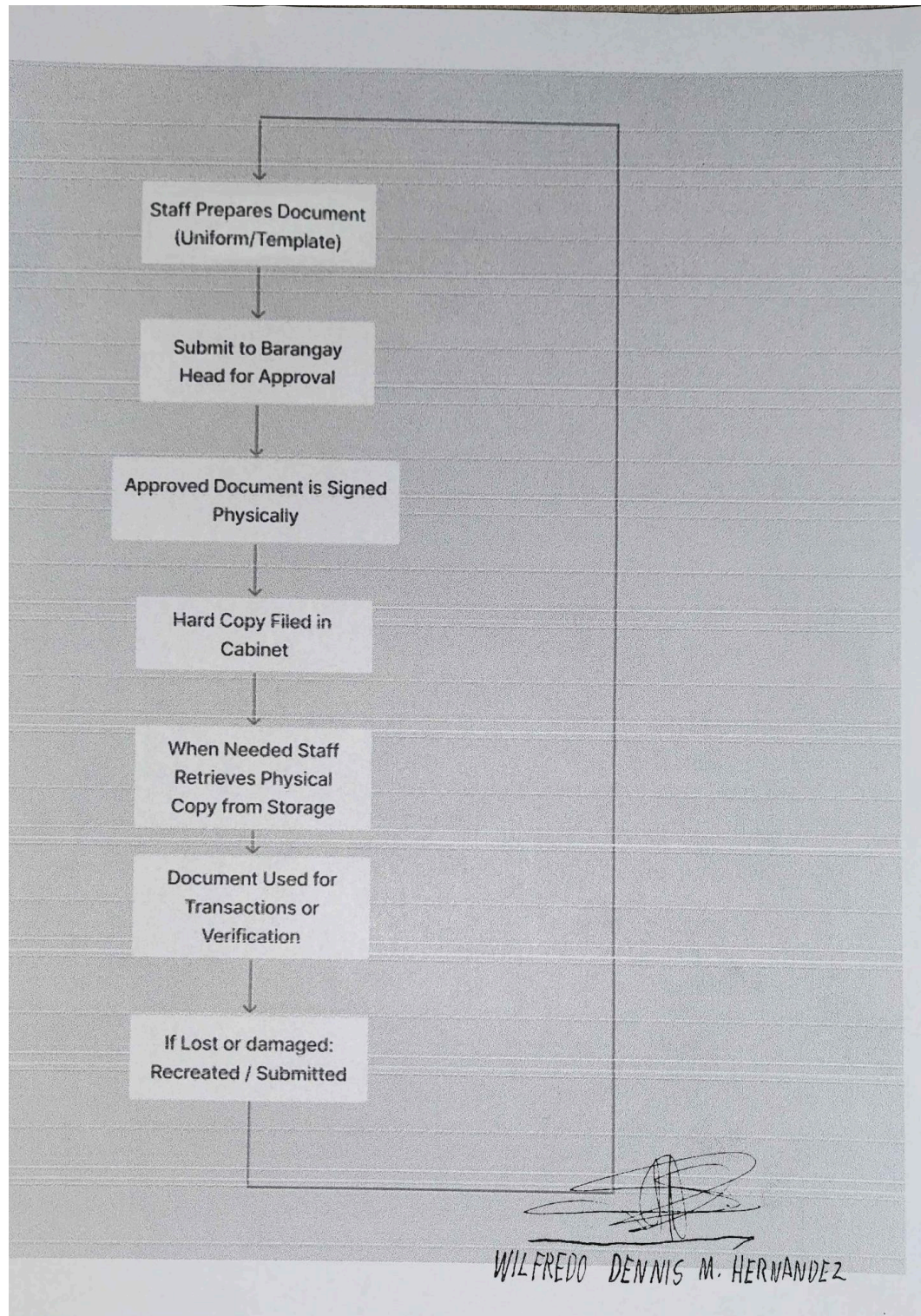
The BAC shall observe the same process and set the new periods according to the same rules followed during the first bidding. After the second failed bidding, however, the BAC may resort to negotiated procurement as provided for in Section 53 of this Act.

Appendix X: Risk Assessment Matrix

Risk ID	Risk Description	Category	Likelihood	Impact	Risk Level	Mitigation / Preventive Measures
R1	Loss or corruption of barangay documents during system migration from manual to digital records.	Data Security	Medium	High	High	Implement regular database backups and use checksum verification to detect corruption. Store backup copies on secure external drives or cloud storage.
R2	Unauthorized access to confidential documents or misuse of admin privileges.	Security / Access Control	Medium	High	High	Enforce Role-Based Access Control (RBAC) , use strong password policies, and restrict access to sensitive files only to verified barangay officials.
R3	Encryption keys lost or exposed due to poor handling or storage.	Cryptographic Risk	Low	High	High	Use a secure key management system; store AES keys in encrypted format accessible only to authorized personnel.
R4	Lack of technical knowledge among barangay staff causing operational delays or misuse.	Human / Operational	High	Medium	High	Conduct user training sessions and provide simple manuals for file uploading, encryption verification, and document retrieval.
R5	Failure of encryption algorithm implementation leading to data inaccessibility.	Technical	Low	High	Medium	Conduct multiple test cases using sample documents before deployment; verify AES-CBC encryption/decryption with SHA-256 hash matching.

R6	System downtime due to server or network failure.	Infrastructure	Medium	Medium	Medium	Utilize a stable hosting platform with regular maintenance; implement automatic reconnection and local caching mechanisms.
R7	Incompatibility with certain file types or document formats used in the barangay.	Functional	Medium	Medium	Medium	Define allowable document formats (e.g., .docx, .pdf, .rtf) and conduct testing for each. Log and handle unsupported uploads gracefully.
R8	User errors such as uploading incorrect or unsigned versions of documents.	Human Error	High	Low	Medium	Add document version control and confirmation prompts before final upload. Include an approval workflow for "draft" and "final" versions.
R9	Non-compliance with RA 9184 documentation or audit requirements.	Legal / Compliance	Low	High	High	Cross-check all uploaded documents with RA 9184 validation forms; ensure system generates transparent audit logs for COA verification.
R10	Data breach from external attackers attempting to decrypt stored files.	Cybersecurity	Low	High	High	Implement multi-layer encryption (AES + SHA-256), secure HTTPS communication, and intrusion detection monitoring.

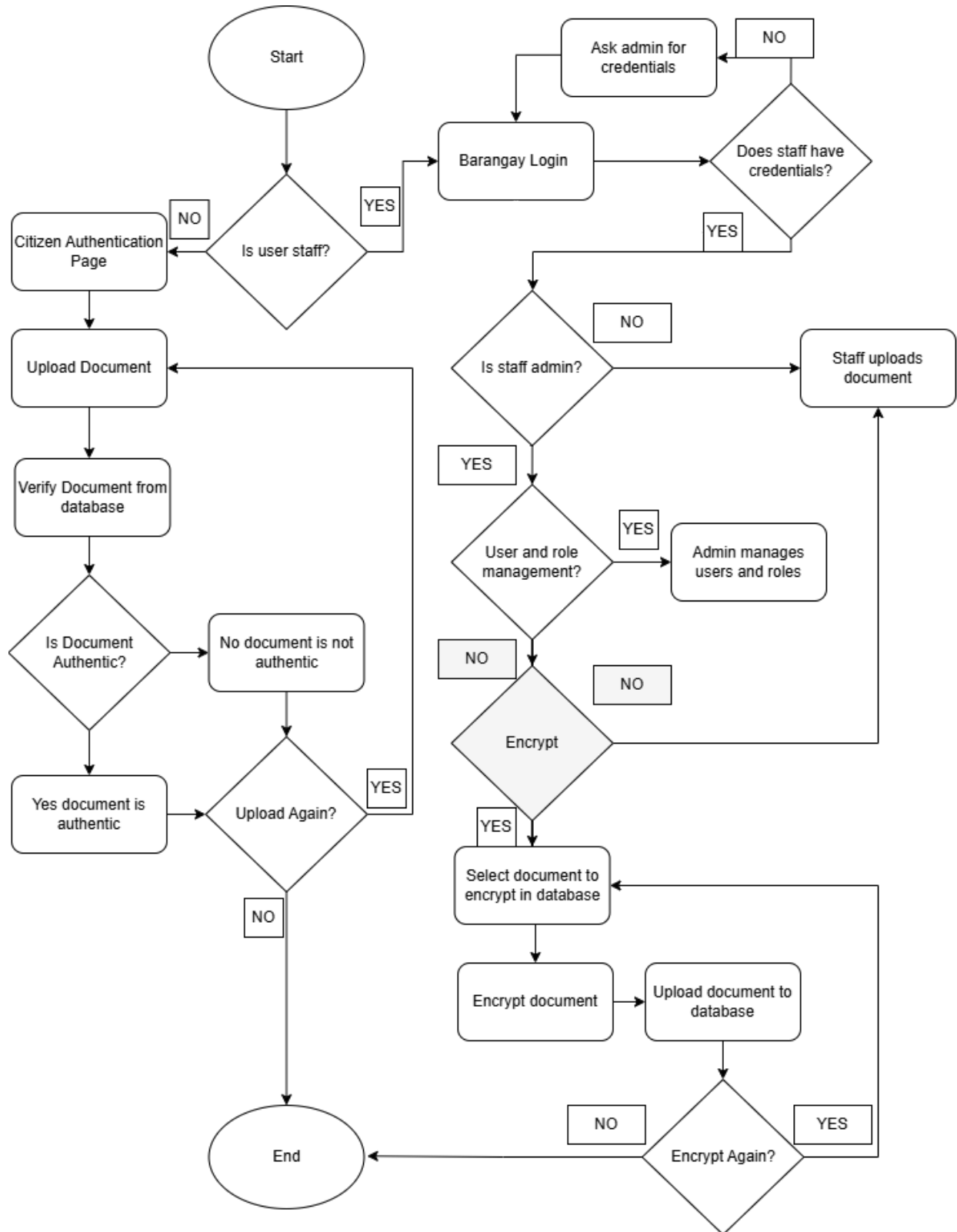
Appendix Y. Current System Workflow Signed



Appendix Z. Gantt Chart

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13
Topic Selection/ Group Formation													
Signed Advisor and students agreement form													
Finding locale for the study													
Formal Interview													
Chapter 2													
Chapter 3													
Chapter 1													
Final Draft													
Proposal Defense													

Appendix AA. Data Flow Chart



CAPSTONE STUDY INTERVIEW QUESTIONS

SECTION A: Procurement Overview

1. How often does the barangay conduct procurement activities (monthly, quarterly, annually)?

A: Annually.

2. Approximately how many procurement-related contracts are produced per year?

A: 10-13 contracts a year

SECTION B: Contract Preparation Process

7. How are contracts prepared?

- Who drafts them?

A: the drafting of contracts and legal documents is generally handled by the following officials and personnel, depending on the nature of the document, it can either come from the secretary or captain

- What software is used (MS Word, Google Docs, etc.)?

A: Usually word ang ginagamit namin through a template

8. Once finalized, how are contracts approved or signed?

A: the contracts are signed ibig sabihin final na, ang magsisign nito usually ay ang barangay captain pati na rin yung supplier

9. Are contracts stored as physical copies, digital copies, or both?

A: Meron kaming dalawang copy both physical saka digital in case magka problema sa physical copy.

SECTION C: Storage and Access

10. Where are digital copies stored?

A: yung mga copya nasa office computer lang

11. Who has access to digital contract files?

A: Barangay officials kagaya ng secretary o captain

12. Is there any version control or tracking mechanism for contract updates?

A: wala kaming version control pero siguro yung filenames at kung kailan ito na modify.

SECTION D: Document Integrity and Accountability

13. How does the barangay ensure that a contract has not been altered after issuance?

A: Gumagamit kami ng official sign documents para maging reference namin, pero medyo mahirap mahanap kung ang document na alter, lalo na kapag ang kontrata ay iilan yung pages.

14. Have there been instances of:

- Lost documents? Minsan meron mga document na namimissplace kapag di nailagay sa storage ng maayos, pero retrievable pa naman siya
- Confusion over versions? Oo lalo na kapag madaming drafts o revisions na ginawa.
- Requests for verification of authenticity? Yes, galing sa contractors or during audits.

15. Is there currently any system used to verify whether a contract is original and unmodified?

A: Wala pa naman.

SECTION E: Public Requests and Transparency

16: How often are procurement documents requested?

A: madalas galing sa contractors para dun sa contracts or supporting documents or coa for audit and compliance checks

17.How are document requests processed (manual retrieval, email, photocopy, etc.)?

A: Kadalasan ibibigay namin yung physical sa nangangailangan, minsan photocopy

SECTION F: Technology Capacity

19. Are staff comfortable using web-based systems?

A: Lahat naman ng staff namin ay merons basic computer skills, pero kung medyo advanced baka hindi.

20. Would a digital verification system for contracts be useful in your operations?

A: oo lalo na sa kapag malayuan ang nangangailangan ng contracts namin.

Appendix AC. Interview Guide Signed

SECTION E: Public Requests and Transparency

16. How often are procurement documents requested?
A: madalas galing sa contractors para dun sa contracts or supporting documents or coa for audit and compliance checks

17. How are document requests processed (manual retrieval, email, photocopy, etc.)?
A: Kadalasan ibibigay namin yung physical sa nangangailangan, minsan photocopy

SECTION F: Technology Capacity

19. Are staff comfortable using web-based systems?
A: Lahat naman ng staff namin ay merons basic computer skills, pero kung medyo advanced baka hindi.

20. Would a digital verification system for contracts be useful in your operations?
A: oo lalo na sa kapag malayuan ang nangangailangan ng contracts namin.

Signature

Appendix AD. Filing Cabinet that contains Important Documents

